

for appropriate α .

[Hint: Prove that $(I(\wp(z + \alpha)))' = \pm 1$, and use the fact that $\wp$ is even.]

4.* Suppose τ is purely imaginary, say $\tau = it$ with $t > 0$. Consider the division of the complex plane into congruent rectangles obtained by considering the lines $x = n/2$, $y = tm/2$ as n and m range over the integers. (An example is the rectangle whose vertices are $0, 1/2, 1/2 + \tau/2$, and $\tau/2$.)

- (a) Show that $\wp$ is real-valued on all these lines, and hence on the boundaries of all these rectangles.
- (b) Prove that $\wp$ maps the interior of each rectangle conformally to the upper (or lower) half-plane.

10 Applications of Theta Functions

The problem of the representation of an integer n as the sum of a given number k of integral squares is one of the most celebrated in the theory of numbers. Its history may be traced back to Diophantus, but begins effectively with Girard's (or Fermat's) theorem that a prime $4m + 1$ is the sum of two squares. Almost every arithmetician of note since Fermat has contributed to the solution of the problem, and it has its puzzles for us still.

G. H. Hardy, 1940

This chapter is devoted to a closer look at the theory of theta functions and some of its applications to combinatorics and number theory.

The theta function is given by the series

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z},$$

which converges for all $z \in \mathbb{C}$, and τ in the upper half-plane.

A remarkable feature of the theta function is its dual nature. When viewed as a function of z , we see it in the arena of elliptic functions, since Θ is periodic with period 1 and “quasi-period” τ . When considered as a function of τ , Θ reveals its modular nature and close connection with the partition function and the problem of representation of integers as sums of squares.

The two main tools allowing us to exploit these links are the triple-product for Θ and its transformation law. Once we have proved these theorems, we give a brief introduction to the connection with partitions, and then pass to proofs of the celebrated theorems about representation of integers as sums of two or four squares.

1 Product formula for the Jacobi theta function

In its most elaborate form, Jacobi's **theta function** is defined for $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ by

$$(1) \quad \Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}.$$

Two significant special cases (or variants) are $\theta(\tau)$ and $\vartheta(t)$, which are defined by

$$\begin{aligned} \theta(\tau) &= \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau}, \quad \tau \in \mathbb{H}, \\ \vartheta(t) &= \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}, \quad t > 0. \end{aligned}$$

In fact, the relation between these various functions is given by $\theta(\tau) = \Theta(0|\tau)$ and $\vartheta(t) = \theta(it)$, with of course, $t > 0$.

We have already encountered these functions several times. For example, in the study of the heat diffusion equation for the circle, in Chapter 4 of Book I, we found that the heat kernel was given by

$$H_t(x) = \sum_{n=-\infty}^{\infty} e^{-4\pi^2 n^2 t} e^{2\pi i n x},$$

and therefore $H_t(x) = \Theta(x|4\pi i t)$.

Another instance was the occurrence of ϑ in the study of the zeta function. In fact, we proved in Chapter 6 that the functional equation of ϑ implied that of ζ , which then led to the analytic continuation of the zeta function.

We begin our closer look at Θ as a function of z , with τ fixed, by recording its basic structural properties, which to a large extent characterize it.

Proposition 1.1 *The function Θ satisfies the following properties:*

- (i) Θ is entire in $z \in \mathbb{C}$ and holomorphic in $\tau \in \mathbb{H}$.
- (ii) $\Theta(z+1|\tau) = \Theta(z|\tau)$.
- (iii) $\Theta(z+\tau|\tau) = \Theta(z|\tau)e^{-\pi i \tau}e^{-2\pi i z}$.
- (iv) $\Theta(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$.

Proof. Suppose that $\text{Im}(\tau) = t \geq t_0 > 0$ and $z = x + iy$ belongs to a bounded set in $\mathbb{C}$, say $|z| \leq M$. Then, the series defining Θ is absolutely and uniformly convergent, since

$$\sum_{n=-\infty}^{\infty} |e^{\pi i n^2 \tau} e^{2\pi i n z}| \leq C \sum_{n \geq 0} e^{-\pi n^2 t_0} e^{2\pi n M} < \infty.$$

Therefore, for each fixed $\tau \in \mathbb{H}$ the function $\Theta(\cdot|\tau)$ is entire, and for each fixed $z \in \mathbb{C}$ the function $\Theta(z|\cdot)$ is holomorphic in the upper half-plane.

Since the exponential $e^{2\pi i n z}$ is periodic of period 1, property (ii) is immediate from the definition of Θ .

To show the third property we may complete the squares in the expression for $\Theta(z + \tau|\tau)$. In detail, we have

$$\begin{aligned} \Theta(z + \tau|\tau) &= \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n(z+\tau)} \\ &= \sum_{n=-\infty}^{\infty} e^{\pi i(n^2+2n)\tau} e^{2\pi i n z} \\ &= \sum_{n=-\infty}^{\infty} e^{\pi i(n+1)^2 \tau} e^{-\pi i \tau} e^{2\pi i n z} \\ &= \sum_{n=-\infty}^{\infty} e^{\pi i(n+1)^2 \tau} e^{-\pi i \tau} e^{2\pi i(n+1)z} e^{-2\pi i z} \\ &= \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}. \end{aligned}$$

Thus we see that $\Theta(z|\tau)$, as a function of z , is periodic with period 1 and “quasi-periodic” with period τ .

To establish the last property it suffices, by what was just shown, to prove that $\Theta(1/2 + \tau/2|\tau) = 0$. Again, we use the interplay between n and n^2 to get

$$\begin{aligned} \Theta(1/2 + \tau/2|\tau) &= \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n(1/2 + \tau/2)} \\ &= \sum_{n=-\infty}^{\infty} (-1)^n e^{\pi i(n^2+n)\tau}. \end{aligned}$$

To see that this last sum is identically zero, it suffices to match $n \geq 0$ with $-n - 1$, and to observe that they have opposite parity, and that $(-n - 1)^2 + (-n - 1) = n^2 + n$. This completes the proof of the proposition.

We consider next a product $\Pi(z|\tau)$ that enjoys the same structural properties as $\Theta(z|\tau)$ as a function of z . This product is defined for $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ by

$$\Pi(z|\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1}e^{2\pi iz})(1 + q^{2n-1}e^{-2\pi iz}),$$

where we have used the notation that is standard in the subject, namely $q = e^{\pi i\tau}$. The function $\Pi(z|\tau)$ is sometimes referred to as the **triple-product**.

Proposition 1.2 *The function $\Pi(z|\tau)$ satisfies the following properties:*

- (i) $\Pi(z, \tau)$ is entire in $z \in \mathbb{C}$ and holomorphic for $\tau \in \mathbb{H}$.
- (ii) $\Pi(z + 1|\tau) = \Pi(z|\tau)$.
- (iii) $\Pi(z + \tau|\tau) = \Pi(z|\tau)e^{-\pi i\tau}e^{-2\pi iz}$.
- (iv) $\Pi(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$. Moreover, these points are simple zeros of $\Pi(\cdot|\tau)$, and $\Pi(\cdot|\tau)$ has no other zeros.

Proof. If $\text{Im}(\tau) = t \geq t_0 > 0$ and $z = x + iy$, then $|q| \leq e^{-\pi t_0} < 1$ and

$$(1 - q^{2n})(1 + q^{2n-1}e^{2\pi iz})(1 + q^{2n-1}e^{-2\pi iz}) = 1 + O(|q|^{2n-1}e^{2\pi|z|}).$$

Since the series $\sum |q|^{2n-1}$ converges, the results for infinite products in Chapter 5 guarantee that $\Pi(z|\tau)$ defines an entire function of z with $\tau \in \mathbb{H}$ fixed, and a holomorphic function for $\tau \in \mathbb{H}$ with $z \in \mathbb{C}$ fixed.

Also, it is clear from the definition that $\Pi(z|\tau)$ is periodic of period 1 in the z variable.

To prove the third property, we first observe that since $q^2 = e^{2\pi i\tau}$ we have

$$\begin{aligned} \Pi(z + \tau|\tau) &= \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1}e^{2\pi i(z+\tau)})(1 + q^{2n-1}e^{-2\pi i(z+\tau)}) \\ &= \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n+1}e^{2\pi iz})(1 + q^{2n-3}e^{-2\pi iz}). \end{aligned}$$

Comparing this last product with $\Pi(z|\tau)$, and isolating the factors that are either missing or extra leads to

$$\Pi(z + \tau|\tau) = \Pi(z|\tau) \left(\frac{1 + q^{-1}e^{-2\pi iz}}{1 + qe^{2\pi iz}} \right).$$

Hence (iii) follows because $(1+x)/(1+x^{-1}) = x$, whenever $x \neq -1$.

Finally, to find the zeros of $\Pi(z|\tau)$ we recall that a product that converges vanishes only if at least one of its factors is zero. Clearly, the factor $(1-q^n)$ never vanishes since $|q| < 1$. The second factor $(1+q^{2n-1}e^{2\pi iz})$ vanishes when $q^{2n-1}e^{2\pi iz} = -1 = e^{\pi i}$. Since $q = e^{\pi i\tau}$, we then have¹

$$(2n-1)\tau + 2z = 1 \pmod{2}.$$

Hence,

$$z = 1/2 + \tau/2 - n\tau \pmod{1},$$

and this takes care of the zeros of the type $1/2 + \tau/2 - n\tau + m$ with $n \geq 1$ and $m \in \mathbb{Z}$. Similarly, the third factor vanishes if

$$(2n-1)\tau - 2z = 1 \pmod{2}$$

which implies that

$$\begin{aligned} z &= -1/2 - \tau/2 + n\tau \pmod{1} \\ &= 1/2 + \tau/2 + n'\tau \pmod{1}, \end{aligned}$$

where $n' \geq 0$. This exhausts the zeros of $\Pi(\cdot|\tau)$. Finally, these zeros are simple, since the function $e^w - 1$ vanishes at the origin to order 1 (a fact obvious from a power series expansion or a simple differentiation).

The importance of the product Π comes from the following theorem, called the product formula for the theta function. The fact that $\Theta(z|\tau)$ and $\Pi(z|\tau)$ satisfy similar properties hints at a close connection between the two. This is indeed the case.

Theorem 1.3 (Product formula) *For all $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ we have the identity $\Theta(z|\tau) = \Pi(z|\tau)$.*

Proof. Fix $\tau \in \mathbb{H}$. We claim first that there exists a constant $c(\tau)$ such that

$$(2) \quad \Theta(z|\tau) = c(\tau)\Pi(z|\tau).$$

In fact, consider the quotient $F(z) = \Theta(z|\tau)/\Pi(z|\tau)$, and note that by the previous two propositions, the function F is entire and doubly periodic with periods 1 and τ . This implies that F is constant as claimed.

¹We use the standard short-hand, $a = b \pmod{c}$, to mean that $a - b$ is an integral multiple of c .

We must now prove that $c(\tau) = 1$ for all τ , and the main point is to establish that $c(\tau) = c(4\tau)$. If we put $z = 1/2$ in (2), so that $e^{2i\pi z} = e^{-2i\pi z} = -1$, we obtain

$$\begin{aligned} \sum_{n=-\infty}^{\infty} (-1)^n q^{n^2} &= c(\tau) \prod_{n=1}^{\infty} (1 - q^{2n})(1 - q^{2n-1})(1 - q^{2n-1}) \\ &= c(\tau) \prod_{n=1}^{\infty} [(1 - q^{2n-1})(1 - q^{2n})] (1 - q^{2n-1}) \\ &= c(\tau) \prod_{n=1}^{\infty} (1 - q^n)(1 - q^{2n-1}). \end{aligned}$$

Hence

$$(3) \quad c(\tau) = \frac{\sum_{n=-\infty}^{\infty} (-1)^n q^{n^2}}{\prod_{n=1}^{\infty} (1 - q^n)(1 - q^{2n-1})}.$$

Next, we put $z = 1/4$ in (2), so that $e^{2i\pi z} = i$. On the one hand, we have

$$\Theta(1/4|\tau) = \sum_{n=-\infty}^{\infty} q^{n^2} i^n,$$

and due to the fact that $1/i = -i$, only the terms corresponding to $n = \text{even} = 2m$ are not cancelled; thus

$$\Theta(1/4|\tau) = \sum_{m=-\infty}^{\infty} q^{4m^2} (-1)^m.$$

On the other hand,

$$\begin{aligned} \Pi(1/4|\tau) &= \prod_{m=1}^{\infty} (1 - q^{2m})(1 + iq^{2m-1})(1 - iq^{2m-1}) \\ &= \prod_{m=1}^{\infty} (1 - q^{2m})(1 + q^{4m-2}) \\ &= \prod_{n=1}^{\infty} (1 - q^{4n})(1 - q^{8n-4}), \end{aligned}$$

where the last line is obtained by considering separately the two cases $2m = 4n - 4$ and $2m = 4n - 2$ in the first factor. Hence

$$(4) \quad c(\tau) = \frac{\sum_{n=-\infty}^{\infty} (-1)^n q^{4n^2}}{\prod_{n=1}^{\infty} (1 - q^{4n})(1 - q^{8n-4})},$$

and combining (3) and (4) establishes our claim that $c(\tau) = c(4\tau)$. Successive applications of this identity give $c(\tau) = c(4^k\tau)$, and since $q^{4^k} = e^{i\pi 4^k \tau} \rightarrow 0$ as $k \rightarrow \infty$, we conclude from (2) that $c(\tau) = 1$. This proves the theorem.

The product formula for the function Θ specializes to its variant $\theta(\tau) = \Theta(0|\tau)$, and this provides a proof that θ is non-vanishing in the upper half-plane.

Corollary 1.4 *If $\text{Im}(\tau) > 0$ and $q = e^{\pi i \tau}$, then*

$$\theta(\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1})^2.$$

Thus $\theta(\tau) \neq 0$ for $\tau \in \mathbb{H}$.

The next corollary shows that the properties of the function Θ now yield the construction of an elliptic function (which is in fact closely related to the Weierstrass $\wp$ function).

Corollary 1.5 *For each fixed $\tau \in \mathbb{H}$, the quotient*

$$(\log \Theta(z|\tau))'' = \frac{\Theta(z|\tau)\Theta''(z|\tau) - (\Theta'(z|\tau))^2}{\Theta(z|\tau)^2}$$

is an elliptic function of order 2 with periods 1 and τ , and with a double pole at $z = 1/2 + \tau/2$.

In the above, the primes ' denote differentiation with respect to the z variable.

Proof. Let $F(z) = (\log \Theta(z|\tau))' = \Theta(z|\tau)' / \Theta(z|\tau)$. Differentiating the identities (ii) and (iii) of Proposition 1.1 gives $F(z+1) = F(z)$, $F(z+\tau) = F(z) - 2\pi i$, and differentiating again shows that $F'(z)$ is doubly periodic. Since $\Theta(z|\tau)$ vanishes only at $z = 1/2 + \tau/2$ in the fundamental parallelogram, the function $F(z)$ has only a single pole, and thus $F'(z)$ has only a double pole there.

The precise connection between $(\log \Theta(z|\tau))''$ and $\wp_\tau(z)$ is stated in Exercise 1.

For an analogy between Θ and the Weierstrass σ function, see Exercise 5 of the previous chapter.

1.1 Further transformation laws

We now come to the study of the transformation relations in the τ -variable, that is, to the modular character of Θ .

Recall that in the previous chapter, the modular character of the Weierstrass $\wp$ function and Eisenstein series E_k was reflected by the two transformations

$$\tau \mapsto \tau + 1 \quad \text{and} \quad \tau \mapsto -1/\tau,$$

which preserve the upper half-plane. In what follows, we shall denote these two transformations by T_1 and S , respectively.

When looking at the Θ function, however, it will be natural to consider instead the transformations

$$T_2 : \tau \mapsto \tau + 2 \quad \text{and} \quad S : \tau \mapsto -1/\tau,$$

since $\Theta(z|\tau + 2) = \Theta(z|\tau)$, but $\Theta(z|\tau + 1) \neq \Theta(z|\tau)$.

Our first task is to study the transformation of $\Theta(z|\tau)$ under the mapping $\tau \mapsto -1/\tau$.

Theorem 1.6 *If $\tau \in \mathbb{H}$, then*

$$(5) \quad \Theta(z|-1/\tau) = \sqrt{\frac{\tau}{i}} e^{\pi i \tau z^2} \Theta(z\tau|\tau) \quad \text{for all } z \in \mathbb{C}.$$

Here $\sqrt{\tau/i}$ denotes the branch of the square root defined on the upper half-plane, that is positive when $\tau = it$, $t > 0$.

Proof. It suffices to prove this formula for $z = x$ real and $\tau = it$ with $t > 0$, since for each fixed $x \in \mathbb{R}$, the two sides of equation (5) are holomorphic functions in the upper half-plane which then agree on the positive imaginary axis, and hence must be equal everywhere. Also, for a fixed $\tau \in \mathbb{H}$ the two sides define holomorphic functions in z that agree on the real axis, and hence must be equal everywhere.

With x real and $\tau = it$ the formula becomes

$$\sum_{n=-\infty}^{\infty} e^{-\pi n^2/t} e^{2\pi i n x} = t^{1/2} e^{-\pi t x^2} \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t} e^{-2\pi n x t}.$$

Replacing x by a , we find that we must prove

$$\sum_{n=-\infty}^{\infty} e^{-\pi t(n+a)^2} = \sum_{n=-\infty}^{\infty} t^{-1/2} e^{-\pi n^2/t} e^{2\pi i n a}.$$

However, this is precisely equation (3) in Chapter 4, which was derived from the Poisson summation formula.

In particular, by setting $z = 0$ in the theorem, we find the following.

Corollary 1.7 *If $\text{Im}(\tau) > 0$, then $\theta(-1/\tau) = \sqrt{\tau/i} \theta(\tau)$.*

Note that if $\tau = it$, then $\theta(\tau) = \vartheta(t)$, and the above relation is precisely the functional equation for ϑ which appeared in Chapter 4.

The transformation law $\theta(-1/\tau) = (\tau/i)^{1/2} \theta(\tau)$ gives us very precise information about the behavior when $\tau \rightarrow 0$. The next corollary will be used later, when we need to analyze the behavior of $\theta(\tau)$ as $\tau \rightarrow 1$.

Corollary 1.8 *If $\tau \in \mathbb{H}$, then*

$$\begin{aligned} \theta(1 - 1/\tau) &= \sqrt{\frac{\tau}{i}} \sum_{n=-\infty}^{\infty} e^{\pi i(n+1/2)^2 \tau} \\ &= \sqrt{\frac{\tau}{i}} (2e^{\pi i \tau/4} + \dots). \end{aligned}$$

The second identity means that $\theta(1 - 1/\tau) \sim \sqrt{\tau/i} 2e^{i\pi\tau/4}$ as $\text{Im}(\tau) \rightarrow \infty$.

Proof. First, we note that n and n^2 have the same parity, so

$$\theta(1 + \tau) = \sum_{n=-\infty}^{\infty} (-1)^n e^{i\pi n^2 \tau} = \Theta(1/2|\tau),$$

hence $\theta(1 - 1/\tau) = \Theta(1/2| -1/\tau)$. Next, we use Theorem 1.6 with $z = 1/2$, and the result is

$$\begin{aligned} \theta(1 - 1/\tau) &= \sqrt{\frac{\tau}{i}} e^{\pi i \tau/4} \Theta(\tau/2|\tau) \\ &= \sqrt{\frac{\tau}{i}} e^{\pi i \tau/4} \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{\pi i n \tau} \\ &= \sqrt{\frac{\tau}{i}} \sum_{n=-\infty}^{\infty} e^{\pi i(n+1/2)^2 \tau}. \end{aligned}$$

The terms corresponding to $n = 0$ and $n = -1$ contribute $2e^{\pi i \tau/4}$, which has absolute value $2e^{-\pi t/4}$ where $\tau = \sigma + it$. Finally, the sum of the other terms $n \neq 0, -1$ is of order

$$O\left(\sum_{k=1}^{\infty} e^{-(k+1/2)^2 \pi t}\right) = O(e^{-9\pi t/4}).$$

Our final corollary of the transformation law pertains to the **Dedekind eta function**, which is defined for $\text{Im}(\tau) > 0$ by

$$\eta(\tau) = e^{\frac{\pi i \tau}{12}} \prod_{n=1}^{\infty} (1 - e^{2\pi i n \tau}).$$

The functional equation for η given below will be relevant to our discussion of the four-square theorem, and in the theory of partitions.

Proposition 1.9 *If $\text{Im}(\tau) > 0$, then $\eta(-1/\tau) = \sqrt{\tau/i} \eta(\tau)$.*

This identity is deduced by differentiating the relation in Theorem 1.6 and evaluating it at $z_0 = 1/2 + \tau/2$. The details are as follows.

Proof. From the product formula for the theta function, we may write with $q = e^{\pi i \tau}$,

$$\Theta(z|\tau) = (1 + qe^{-2\pi iz}) \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1}e^{2\pi iz})(1 + q^{2n+1}e^{-2\pi iz}),$$

and since the first factor vanishes at $z_0 = 1/2 + \tau/2$, we see that

$$\Theta'(z_0|\tau) = 2\pi i H(\tau), \quad \text{where } H(\tau) = \prod_{n=1}^{\infty} (1 - e^{2\pi i n \tau})^3.$$

Next, we observe that with $-1/\tau$ replaced by τ in (5), we obtain

$$\Theta(z|\tau) = \sqrt{i/\tau} e^{-\pi iz^2/\tau} \Theta(-z/\tau | -1/\tau).$$

If we differentiate this expression and then evaluate it at the point $z_0 = 1/2 + \tau/2$, we find

$$2\pi i H(\tau) = \sqrt{i/\tau} e^{-\frac{\pi i}{4\tau}} e^{-\frac{\pi i}{2}} e^{-\frac{\pi i \tau}{4}} \left(\frac{-2\pi i}{\tau} \right) H(-1/\tau).$$

Hence

$$e^{\frac{\pi i \tau}{4}} H(\tau) = \left(\frac{i}{\tau} \right)^{3/2} e^{-\frac{\pi i}{4\tau}} H(-1/\tau).$$

We note that when $\tau = it$, with $t > 0$, the function $\eta(\tau)$ is positive, and thus taking the cube root of the above gives $\eta(\tau) = \sqrt{i/\tau} \eta(-1/\tau)$; therefore this identity holds for all $\tau \in \mathbb{H}$ by analytic continuation.

A connection between the function η and the theory of elliptic functions is given in Problem 5.

2 Generating functions

Given a sequence $\{F_n\}_{n=0}^\infty$, which may arise either combinatorially, recursively, or in terms of some number-theoretic law, an important tool in its study is the passage to its **generating function**, defined by

$$F(x) = \sum_{n=0}^{\infty} F_n x^n.$$

Often times, the defining properties of the sequence $\{F_n\}$ imply interesting algebraic or analytic properties of the function $F(x)$, and exploiting these can eventually lead us back to new insights about the sequence $\{F_n\}$. A very simple-minded example is given by the Fibonacci sequence. (See Exercise 2). Here we want to study less elementary examples of this idea, related to the Θ function.

We shall first discuss very briefly the theory of partitions.

The **partition function** is defined as follows: if n is a positive integer, we let $p(n)$ denote the numbers of ways n can be written as a sum of positive integers. For instance, $p(1) = 1$, and $p(2) = 2$ since $2 = 2 + 0 = 1 + 1$. Also, $p(3) = 3$ since $3 = 3 + 0 = 2 + 1 = 1 + 1 + 1$. We set $p(0) = 1$ and collect some further values of $p(n)$ in the following table.

n	0	1	2	3	4	5	6	7	8	$\cdots$	12
$p(n)$	1	1	2	3	5	7	11	15	22	$\cdots$	77

The first theorem is Euler's identity for the generating function of the partition sequence $\{p(n)\}$, which is reminiscent of the product formula for the zeta function.

Theorem 2.1 *If $|x| < 1$, then $\sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}$.*

Formally, we can write each fraction as

$$\frac{1}{1-x^k} = \sum_{m=0}^{\infty} x^{km},$$

and multiply these out together to obtain $p(n)$ as the coefficient of x^n . Indeed, when we group together equal integers in a partition of n , this partition can be written as

$$n = m_1 k_1 + \cdots + m_r k_r,$$

where $k_1, \dots, k_r$ are distinct positive integers. This partition corresponds to the term

$$(x^{k_1})^{m_1} \dots (x^{k_r})^{m_r}$$

that arises in the product.

The justification of this formal argument proceeds as in the proof of the product formula for the zeta function (Section 1, Chapter 7); this is based on the convergence of the product $\prod 1/(1 - x^k)$. This convergence in turn follows from the fact that for each fixed $|x| < 1$ one has

$$\frac{1}{1 - x^k} = 1 + O(x^k).$$

A similar argument shows that the product $\prod 1/(1 - x^{2n-1})$ is equal to the generating function for $p_o(n)$, the number of partitions of n into odd parts. Also, $\prod (1 + x^n)$ is the generating function for $p_u(n)$, the number of partitions of n into unequal parts. Remarkably, $p_o(n) = p_u(n)$ for all n , and this translates into the identity

$$\prod_{n=1}^{\infty} \left(\frac{1}{1 - x^{2n-1}} \right) = \prod_{n=1}^{\infty} (1 + x^n).$$

To prove this note that $(1 + x^n)(1 - x^n) = 1 - x^{2n}$, and therefore

$$\prod_{n=1}^{\infty} (1 + x^n) \prod_{n=1}^{\infty} (1 - x^n) = \prod_{n=1}^{\infty} (1 - x^{2n}).$$

Moreover, taking into account the parity of integers, we have

$$\prod_{n=1}^{\infty} (1 - x^{2n}) \prod_{n=1}^{\infty} (1 - x^{2n-1}) = \prod_{n=1}^{\infty} (1 - x^n),$$

which combined with the above proves the desired identity.

The proposition that follows is deeper, and in fact involves the Θ function directly. Let $p_{e,u}(n)$ denote the number of partitions of n into an even number of unequal parts, and $p_{o,u}(n)$ the number of partitions of n into an odd number of unequal parts. Then, Euler proved that, unless n is a pentagonal number, one has $p_{e,u}(n) = p_{o,u}(n)$. By definition, **pentagonal numbers**² are integers n of the form $k(3k + 1)/2$, with $k \in \mathbb{Z}$. For

²The traditional definition is as follows. Integers of the form $n = k(k - 1)/2$, $k \in \mathbb{Z}$, are “triangular numbers”; those of the form $n = k^2$ are “squares”; and those of the form $k(3k + 1)/2$ are “pentagonal numbers.” In general, numbers of the form $(k/2)((\ell - 2)k + \ell - 4)$ are associated with an ℓ -sided polygon.

example, the first few pentagonal numbers are 1, 2, 5, 7, 12, 15, 22, 26, ... In fact, if n is pentagonal, then

$$p_{e,u}(n) - p_{o,u}(n) = (-1)^k, \quad \text{if } n = k(3k+1)/2.$$

To prove this result, we first observe that

$$\prod_{n=1}^{\infty} (1 - x^n) = \sum_{n=1}^{\infty} [p_{e,u}(n) - p_{o,u}(n)] x^n.$$

This follows since multiplying the terms in the product, we obtain terms of the form $(-1)^r x^{n_1 + \dots + n_r}$ where the integers $n_1, \dots, n_r$ are distinct. Hence in the coefficient of x^n , each partition $n_1 + \dots + n_r$ of n into an even number of unequal parts contributes for +1 (r is even), and each partition into an odd number of unequal parts contributes -1 (r is odd). This gives precisely the coefficient $p_{e,u}(n) - p_{o,u}(n)$.

With the above identity, we see that Euler's theorem is a consequence of the following proposition.

Proposition 2.2
$$\prod_{n=1}^{\infty} (1 - x^n) = \sum_{k=-\infty}^{\infty} (-1)^k x^{\frac{k(3k+1)}{2}}.$$

Proof. If we set $x = e^{2\pi i u}$, then we can write

$$\prod_{n=1}^{\infty} (1 - x^n) = \prod_{n=1}^{\infty} (1 - e^{2\pi i n u})$$

in terms of the triple product

$$\prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1} e^{2\pi i z})(1 + q^{2n-1} e^{-2\pi i z})$$

by letting $q = e^{3\pi i u}$ and $z = 1/2 + u/2$. This is because

$$\prod_{n=1}^{\infty} (1 - e^{2\pi i 3nu})(1 - e^{2\pi i (3n-1)u})(1 - e^{2\pi i (3n-2)u}) = \prod_{n=1}^{\infty} (1 - e^{2\pi i nu}).$$

By Theorem 1.3 the product equals

$$\sum_{n=-\infty}^{\infty} e^{3\pi i n^2 u} (-1)^n e^{2\pi i n u/2} = \sum_{n=-\infty}^{\infty} (-1)^n e^{\pi i n(3n+1)u}$$

$$= \sum_{n=-\infty}^{\infty} (-1)^n x^{n(3n+1)/2},$$

which was to be proved.

We make a final comment about the partition function $p(n)$. The nature of its growth as $n \rightarrow \infty$ can be analyzed in terms of the behavior of $1/\prod_{n=1}^{\infty} (1-x)^n$ as $|x| \rightarrow 1$. In fact, by elementary considerations, we can get a rough order of growth of $p(n)$ from the growth of the generating function as $x \rightarrow 1$; see Exercises 5 and 6. A more refined analysis requires the transformation properties of the generating function which goes back to the corresponding Proposition 1.9 about η . This leads to a very good asymptotic formula for $p(n)$. It may be found in Appendix A.

3 The theorems about sums of squares

The ancient Greeks were fascinated by triples of integers (a, b, c) that occurred as sides of right triangles. These are the “Pythagorean triples,” which satisfy $a^2 + b^2 = c^2$. According to Diophantus of Alexandria (ca. 250 AD), if c is an integer of the above kind, and a and b have no common factors (a case to which one may easily reduce), then c is the sum of two squares, that is, $c = m^2 + n^2$ with $m, n \in \mathbb{Z}$; and conversely, any such c arises as the hypotenuse of a triangle whose sides are given by a Pythagorean triple (a, b, c) . (See Exercise 8.) Therefore, it is natural to ask the following question: which integers can be written as the sum of two squares? It is easy to see that no number of the form $4k + 3$ can be so written, but to determine which integers can be expressed in this way is not obvious.

Let us pose the question in a more quantitative form. We define $r_2(n)$ to be the number of ways n can be written as the sum of two squares, counting obvious repetitions; that is, $r_2(n)$ is the number of pairs (x, y) , $x, y \in \mathbb{Z}$, so that

$$n = x^2 + y^2.$$

For example, $r_2(3) = 0$, but $r_2(5) = 8$ because $5 = (\pm 2)^2 + (\pm 1)^2$, and also $5 = (\pm 1)^2 + (\pm 2)^2$. Hence, our first problem can be posed as follows:

Sum of two squares: Which integers can be written as a sum of two squares? More precisely, can one determine an expression for $r_2(n)$?

Next, since not every positive integer can be expressed as the sum of two squares, we may ask if three squares, or possibly four squares suffice.

However, the fact is that there are infinitely many integers that cannot be written as the sum of three squares, since it is easy to check that no integer of the form $8k + 7$ can be so written. So we turn to the question of four squares and define, in analogy with $r_2(n)$, the function $r_4(n)$ to be the number of ways of expressing n as a sum of four squares. Therefore, a second problem that arises is:

Sum of four squares: Can every positive integer be written as a sum of four squares? More precisely, determine a formula for $r_4(n)$.

It turns out that the problems of two squares and four squares, which go back to the third century, were not resolved until about 1500 years later, and their full solution was first given by the use of Jacobi's theory of theta functions!

3.1 The two-squares theorem

The problem of representing an integer as the sum of two squares, while obviously additive in nature, has a nice multiplicative aspect: if n and m are two integers that can be written as the sum of two squares, then so can their product nm . Indeed, suppose $n = a^2 + b^2$, $m = c^2 + d^2$, and consider the complex number

$$x + iy = (a + ib)(c + id).$$

Clearly, x and y are integers since $a, b, c, d \in \mathbb{Z}$, and by taking absolute values on both sides we see that

$$x^2 + y^2 = (a^2 + b^2)(c^2 + d^2),$$

and it follows that $nm = x^2 + y^2$.

For these reasons the divisibility properties of n play a crucial role in determining $r_2(n)$. To state the basic result we define two new **divisor functions**: we let $d_1(n)$ denote the number of divisors of n of the form $4k + 1$, and $d_3(n)$ the number of divisors of n of the form $4k + 3$. The main result of this section provides a complete answer to the two-squares problem:

Theorem 3.1 *If $n \geq 1$, then $r_2(n) = 4(d_1(n) - d_3(n))$.*

A direct consequence of the above formula for $r_2(n)$ may be stated as follows. If $n = p_1^{a_1} \cdots p_r^{a_r}$ is the prime factorization of n where $p_1, \dots, p_r$ are distinct, then:

The positive integer n can be represented as the sum of two squares if and only if every prime p_j of the form $4k + 3$ that occurs in the factorization of n has an even exponent a_j .

The proof of this deduction is outlined in Exercise 9.

To prove the theorem, we first establish a crucial relationship that identifies the generating function of the sequence $\{r_2(n)\}_{n=1}^{\infty}$ with the square of the θ function, namely

$$(6) \quad \theta(\tau)^2 = \sum_{n=0}^{\infty} r_2(n) q^n,$$

whenever $q = e^{\pi i \tau}$ with $\tau \in \mathbb{H}$. The proof of this identity relies simply on the definition of r_2 and θ . Indeed, if we first recall that $\theta(\tau) = \sum_{n=-\infty}^{\infty} q^{n^2}$, then we obtain

$$\begin{aligned} \theta(\tau)^2 &= \left(\sum_{n_1=-\infty}^{\infty} q^{n_1^2} \right) \left(\sum_{n_2=-\infty}^{\infty} q^{n_2^2} \right) \\ &= \sum_{(n_1, n_2) \in \mathbb{Z} \times \mathbb{Z}} q^{n_1^2 + n_2^2} \\ &= \sum_{n=0}^{\infty} r_2(n) q^n, \end{aligned}$$

since $r_2(n)$ counts the number of pairs (n_1, n_2) with $n_1^2 + n_2^2 = n$.

Proposition 3.2 *The identity $r_2(n) = 4(d_1(n) - d_3(n))$, $n \geq 1$, is equivalent to the identities*

$$(7) \quad \theta(\tau)^2 = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = 1 + 4 \sum_{n=1}^{\infty} \frac{q^n}{1 + q^{2n}},$$

whenever $q = e^{\pi i \tau}$ and $\tau \in \mathbb{H}$.

Proof. We note first that both series converge absolutely since $|q| < 1$, and the first equals the second, because $1/(q^n + q^{-n}) = q^{|n|}/(1 + q^{2|n|})$.

Since $(1 + q^{2n})^{-1} = (1 - q^{2n})/(1 - q^{4n})$, the right-hand side of (7) equals

$$1 + 4 \sum_{n=1}^{\infty} \left(\frac{q^n}{1 - q^{4n}} - \frac{q^{3n}}{1 - q^{4n}} \right).$$

However, since $1/(1 - q^{4n}) = \sum_{m=0}^{\infty} q^{4nm}$, we have

$$\sum_{n=1}^{\infty} \frac{q^n}{1 - q^{4n}} = \sum_{n=1}^{\infty} \sum_{m=0}^{\infty} q^{n(4m+1)} = \sum_{k=1}^{\infty} d_1(k) q^k,$$

because $d_1(k)$ counts the number of divisors of k that are of the form $4m + 1$. Observe that the series $\sum d_1(k) q^k$ converges since $d_1(k) \leq k$.

A similar argument shows that

$$\sum_{n=1}^{\infty} \frac{q^{3n}}{1 - q^{4n}} = \sum_{k=1}^{\infty} d_3(k) q^k,$$

and the proof of the proposition is complete.

In effect, we see that the identity (6) links the original problem in arithmetic with the problem in complex analysis of establishing the relation (7).

We shall now find it convenient to use $\mathcal{C}(\tau)$ to denote³

$$(8) \quad \mathcal{C}(\tau) = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = \sum_{n=-\infty}^{\infty} \frac{1}{\cos(n\pi\tau)},$$

where $q = e^{\pi i \tau}$ and $\tau \in \mathbb{H}$. Our work then becomes to prove the identity $\theta(\tau)^2 = \mathcal{C}(\tau)$.

What is truly remarkable are the different yet parallel ways that the functions θ and $\mathcal{C}$ arise. The genesis of the function θ may be thought to be the heat diffusion equation on the real line; the corresponding heat kernel is given in terms of the Gaussian $e^{-\pi x^2}$ which is its own Fourier transform; and finally the transformation rule for θ results from the Poisson summation formula.

The parallel with $\mathcal{C}$ is that it arises from another differential equation: the steady-state heat equation in a strip; there, the corresponding kernel is $1/\cosh \pi x$ (Section 1.3, Chapter 8), which again is its own Fourier transform (Example 3, Chapter 3). The transformation rule for $\mathcal{C}$ results, once again, from the Poisson summation formula.

To prove the identity $\theta^2 = \mathcal{C}$ we will first show that these two functions satisfy the same structural properties. For θ^2 we had the transformation law $\theta(\tau)^2 = (i/\tau)\theta(-1/\tau)^2$ (Corollary 1.7).

³We denote the function by $\mathcal{C}$ because we are summing a series of cosines.

An identical transformation law holds for $\mathcal{C}(\tau)$! Indeed, if we set $a = 0$ in the relation (5) of Chapter 4 we obtain

$$\sum_{n=-\infty}^{\infty} \frac{1}{\cosh(\pi n t)} = \frac{1}{t} \sum_{n=-\infty}^{\infty} \frac{1}{\cosh(\pi n/t)}.$$

This is precisely the identity

$$\mathcal{C}(\tau) = (i/\tau) \mathcal{C}(-1/\tau)$$

for $\tau = it$, $t > 0$, which therefore also holds for all $\tau \in \mathbb{H}$ by analytic continuation.

It is also obvious from their definitions that both $\theta(\tau)^2$ and $\mathcal{C}(\tau)$ tend to 1 as $\text{Im}(\tau) \rightarrow \infty$. The last property we want to examine is the behavior of both functions at the “cusp” $\tau = 1$.⁴

For θ^2 we shall invoke Corollary 1.8 to see that $\theta(1 - 1/\tau)^2 \sim 4(\tau/i)e^{\pi i\tau/2}$ as $\text{Im}(\tau) \rightarrow \infty$.

For $\mathcal{C}$ we can do the same, again using the Poisson summation formula. In fact, if we set $a = 1/2$ in equation (5), Chapter 4, we find

$$\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{\cosh(\pi n/t)} = t \sum_{n=-\infty}^{\infty} \frac{1}{\cosh(\pi(n + 1/2)t)}.$$

Therefore, by analytic continuation we deduce that

$$\mathcal{C}(1 - 1/\tau) = \left(\frac{\tau}{i}\right) \sum_{n=-\infty}^{\infty} \frac{1}{\cosh(\pi(n + 1/2)\tau)}.$$

The main terms of this sum are those for $n = -1$ and $n = 0$. This easily gives

$$\mathcal{C}(1 - 1/\tau) = 4 \left(\frac{\tau}{i}\right) e^{\pi i\tau/2} + O(|\tau|e^{-3\pi t/2}), \quad \text{as } t \rightarrow \infty,$$

and where $\tau = \sigma + it$. We summarize our conclusions in a proposition.

Proposition 3.3 *The function $\mathcal{C}(\tau) = \sum 1/\cosh(\pi n\tau)$, defined in the upper half-plane, satisfies*

- (i) $\mathcal{C}(\tau + 2) = \mathcal{C}(\tau)$.
- (ii) $\mathcal{C}(\tau) = (i/\tau)\mathcal{C}(-1/\tau)$.

⁴Why we refer to the point $\tau = 1$ as a cusp, and the reason for its importance, will become clear later on.

(iii) $\mathcal{C}(\tau) \rightarrow 1$ as $\text{Im}(\tau) \rightarrow \infty$.

(iv) $\mathcal{C}(1 - 1/\tau) \sim 4(\tau/i)e^{\pi i\tau/2}$ as $\text{Im}(\tau) \rightarrow \infty$.

Moreover, $\theta(\tau)^2$ satisfies the same properties.

With this proposition, we prove the identity of $\theta(\tau)^2 = \mathcal{C}(\tau)$ with the aid of the following theorem, in which we shall ultimately set $f = \mathcal{C}/\theta^2$.

Theorem 3.4 *Suppose f is a holomorphic function in the upper half-plane that satisfies:*

(i) $f(\tau + 2) = f(\tau)$,

(ii) $f(-1/\tau) = f(\tau)$,

(iii) $f(\tau)$ is bounded,

then f is constant.

For the proof of this theorem, we introduce the following subset of the closed upper half-plane, which is defined by

$$\mathcal{F} = \{\tau \in \overline{\mathbb{H}} : |\text{Re}(\tau)| \leq 1 \text{ and } |\tau| \geq 1\},$$

and illustrated in Figure 1.

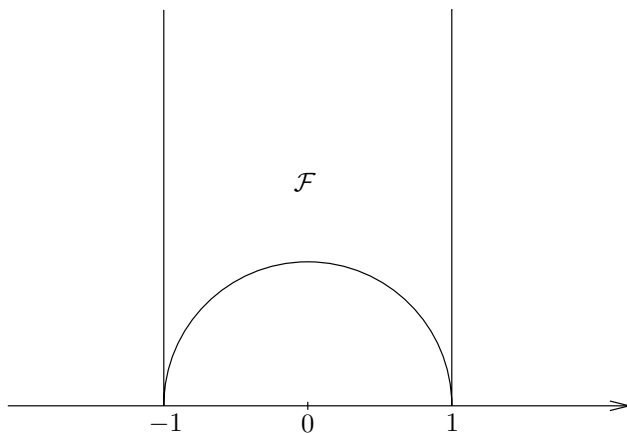


Figure 1. The domain $\mathcal{F}$

The points corresponding to $\tau = \pm 1$ are called **cusps**. They are equivalent under the mapping $\tau \mapsto \tau + 2$.

Lemma 3.5 *Every point in the upper half-plane can be mapped into $\mathcal{F}$ using repeatedly one or another of the following fractional linear transformations or their inverses:*

$$T_2 : \tau \mapsto \tau + 2, \quad S : \tau \mapsto -1/\tau.$$

For this reason, $\mathcal{F}$ is called the **fundamental domain**⁵ for the group of transformations generated by T_2 and S .

In fact, we let G denote the group generated by T_2 and S . Since T_2 and S are fractional linear transformations, we may represent an element $g \in G$ by a matrix

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

with the understanding that

$$g(\tau) = \frac{a\tau + b}{c\tau + d}.$$

Since the matrices representing T_2 and S have integer coefficients and determinant 1, the same is true for all matrices of elements in G . In particular, if $\tau \in \mathbb{H}$, then

$$(9) \quad \operatorname{Im}(g(\tau)) = \frac{\operatorname{Im}(\tau)}{|c\tau + d|^2}.$$

Proof of Lemma 3.5. Let $\tau \in \mathbb{H}$. If $g \in G$ with $g(\tau) = (a\tau + b)/(c\tau + d)$, then c and d are integers, and by (9) we may choose a $g_0 \in G$ such that $\operatorname{Im}(g_0(\tau))$ is maximal. Since the translations T_2 and their inverses do not change imaginary parts, we may apply finitely many of them to see that there exists $g_1 \in G$ with $|\operatorname{Re}(g_1(\tau))| \leq 1$ and $\operatorname{Im}(g_1(\tau))$ is maximal. It now suffices to prove that $|g_1(\tau)| \geq 1$ to conclude that $g_1(\tau) \in \mathcal{F}$. If this were not true, that is, $|g_1(\tau)| < 1$, then $\operatorname{Im}(Sg_1(\tau))$ would be greater than $\operatorname{Im}(g_1(\tau))$ since

$$\operatorname{Im}(Sg_1(\tau)) = \operatorname{Im}(-1/g_1(\tau)) = -\frac{\operatorname{Im}(\overline{g_1(\tau)})}{|g_1(\tau)|^2} > \operatorname{Im}(g_1(\tau)),$$

and this contradicts the maximality of $\operatorname{Im}(g_1(\tau))$.

We can now prove the theorem. Suppose f is not constant, and let $g(z) = f(\tau)$ where $z = e^{\pi i \tau}$. The function g is well defined for z in the

⁵Strictly speaking, the notion of a fundamental domain requires that every point have a unique representative in the domain. In the present case, ambiguity arises only for points that are on the boundary of $\mathcal{F}$.

punctured unit disc, since f is periodic of period 2, and moreover, g is bounded near the origin by assumption (iii) of the theorem. Hence 0 is a removable singularity for g , and $\lim_{z \rightarrow 0} g(z) = \lim_{\text{Im}(\tau) \rightarrow \infty} f(\tau)$ exists. So by the maximum modulus principle,

$$\lim_{\text{Im}(\tau) \rightarrow \infty} |f(\tau)| < \sup_{\tau \in \mathcal{F}} |f(\tau)|.$$

Now we must investigate the behavior of f at the points $\tau = \pm 1$. Since $f(\tau + 2) = f(\tau)$, it suffices to consider the point $\tau = 1$. We claim that

$$\lim_{\text{Im}(\tau) \rightarrow \infty} f(1 - 1/\tau)$$

exists and moreover

$$\lim_{\text{Im}(\tau) \rightarrow \infty} |f(1 - 1/\tau)| < \sup_{\tau \in \mathcal{F}} |f(\tau)|.$$

The argument is essentially the same as the one above, except that we first need to interchange $\tau = 1$ with the point at infinity. In other words, we wish to investigate the behavior of $F(\tau) = f(1 - 1/\tau)$ for τ near ∞ . The important step is to prove that F is periodic. To this end, we consider the fractional linear transformation associated to the matrix

$$U_n = \begin{pmatrix} 1 - n & n \\ -n & 1 + n \end{pmatrix},$$

that is,

$$\tau \mapsto \frac{(1 - n)\tau + n}{-n\tau + (1 + n)},$$

which maps 1 to 1. Now let $\mu(\tau) = 1/(1 - \tau)$ which maps 1 to ∞ , and whose inverse $\mu^{-1}(\tau) = 1 - 1/\tau$ takes ∞ to 1. Then

$$U_n = \mu^{-1} T_n \mu,$$

where T_n is the translation $T_n(\tau) = \tau + n$. As a consequence,

$$U_n U_m = U_{n+m},$$

and

$$U_{-1} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} = T_2 S.$$

Thus any U_n can be obtained by finitely many applications of T_2 , S , or their inverses. Since f is invariant under T_2 and S , it is also invariant under U_m . So we find that

$$f(\mu^{-1}T_n\mu(\tau)) = f(\tau).$$

Therefore, if we let $F(\tau) = f(\mu^{-1}(\tau)) = f(1 - 1/\tau)$, we find that F is periodic of period 1, that is,

$$F(T_n\tau) = F(\tau) \quad \text{for every integer } n.$$

Now, by the previous argument, if we set $h(z) = F(\tau)$ with $z = e^{2\pi i\tau}$, we see that h has a removable singularity at $z = 0$, and the desired inequality follows by the maximum principle.

We conclude from this analysis that f attains its maximum in the interior of the upper half-plane, and this contradicts the maximum principle.

The proof of the two-squares theorem is now only one step away.

We consider the function $f(\tau) = \mathcal{C}(\tau)/\theta(\tau)^2$. Since we know by the product formula that $\theta(\tau)$ does not vanish in the upper half-plane (Corollary 1.4), we find that f is holomorphic in $\mathbb{H}$. Moreover, by Proposition 3.3, f is invariant under the transformations T_2 and S , that is, $f(\tau + 2) = f(\tau)$ and $f(-1/\tau) = f(\tau)$. Finally, in the fundamental domain $\mathcal{F}$, the function $f(\tau)$ is bounded, and in fact tends to 1 as $\text{Im}(\tau)$ tends to infinity, or as τ tends to the cusps ± 1 . This is because of properties (iii) and (iv) in Proposition 3.3, which are verified by both $\mathcal{C}$ and θ^2 . Thus f is bounded in $\mathbb{H}$. The result is that f is a constant, which must be 1, proving that $\theta(\tau)^2 = \mathcal{C}(\tau)$, and with it the two-squares theorem.

3.2 The four-squares theorem

Statement of the theorem

In the rest of this chapter, we shall consider the case of four squares. More precisely, we will prove that every positive integer is the sum of four squares, and moreover we will determine a formula for $r_4(n)$ that describes the number of ways this can be done.

We need to introduce another divisor function, which we denote by $\sigma_1^*(n)$, and which equals the sum of divisors of n that are not divisible by 4. The main theorem we shall prove is the following.

Theorem 3.6 *Every positive integer is the sum of four squares, and moreover*

$$r_4(n) = 8\sigma_1^*(n) \quad \text{for all } n \geq 1.$$

As before, we relate the sequence $\{r_4(n)\}$ via its generating function to an appropriate power of the function θ , which in this case is its fourth power. The result is that

$$\theta(\tau)^4 = \sum_{n=0}^{\infty} r_4(n) q^n$$

whenever $q = e^{\pi i \tau}$ with $\tau \in \mathbb{H}$.

The next step is to find the modular function whose equality with $\theta(\tau)^4$ expresses the identity $r_4(n) = 8\sigma_1^*(n)$. Unfortunately, here there is nothing as simple as the function $\mathcal{C}(\tau)$ that arose in the two-squares theorem; instead we shall need to construct a rather subtle variant of the Eisenstein series considered in the previous chapter. In fact, we define

$$E_2^*(\tau) = \sum_m \sum_n \frac{1}{\left(\frac{m\tau}{2} + n\right)^2} - \sum_m \sum_n \frac{1}{\left(m\tau + \frac{n}{2}\right)^2}$$

for $\tau \in \mathbb{H}$. The indicated order of summation is critical, since the above series do not converge absolutely. The following reduces the four-squares theorem to the modular properties of E_2^* .

Proposition 3.7 *The assertion $r_4(n) = 8\sigma_1^*(n)$ is equivalent to the identity*

$$\theta(\tau)^4 = \frac{-1}{\pi^2} E_2^*(\tau), \quad \text{where } \tau \in \mathbb{H}.$$

Proof. It suffices to prove that if $q = e^{\pi i \tau}$, then

$$\frac{-1}{\pi^2} E_2^*(\tau) = 1 + \sum_{k=1}^{\infty} 8\sigma_1^*(k) q^k.$$

First, recall the forbidden Eisenstein series that we considered in the last section of the previous chapter, and which is defined by

$$F(\tau) = \sum_m \left[\sum_n \frac{1}{(m\tau + n)^2} \right],$$

where the term $n = m = 0$ is omitted. Since the sum above is not absolutely convergent, the order of summation, first in n and then in m , is crucial. With this in mind, the definitions of E_2^* and F give immediately

$$(10) \quad E_2^*(\tau) = F\left(\frac{\tau}{2}\right) - 4F(2\tau).$$

In Corollary 2.6 (and Exercise 7) of the last chapter, we proved that

$$F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum_{k=1}^{\infty} \sigma_1(k) e^{2\pi i k \tau},$$

where $\sigma_1(k)$ is the sum of the divisors of k .

Now observe that

$$\sigma_1^*(n) = \begin{cases} \sigma_1(n) & \text{if } n \text{ is not divisible by } 4, \\ \sigma_1(n) - 4\sigma_1(n/4) & \text{if } n \text{ is divisible by } 4. \end{cases}$$

Indeed, if n is not divisible by 4, then no divisors of n are divisible by 4. If $n = 4\tilde{n}$, and d is a divisor of n that is divisible by 4, say $d = 4\tilde{d}$, then $\tilde{d}$ divides $\tilde{n}$. This gives the second formula. Therefore, from this observation and (10) we find that

$$E_2^*(\tau) = -\pi^2 - 8\pi^2 \sum_{k=1}^{\infty} \sigma_1^*(k) e^{\pi i k \tau},$$

and the proof of the proposition is complete.

We have therefore reduced Theorem 3.6 to the identity $\theta^4 = -\pi^{-2}E_2^*$, and the key to establish this relation is that E_2^* satisfies the same modular properties as $\theta(\tau)^4$.

Proposition 3.8 *The function $E_2^*(\tau)$ defined in the upper half-plane has the following properties:*

- (i) $E_2^*(\tau + 2) = E_2^*(\tau)$.
- (ii) $E_2^*(\tau) = -\tau^{-2}E_2^*(-1/\tau)$.
- (iii) $E_2^*(\tau) \rightarrow -\pi^2$ as $\text{Im}(\tau) \rightarrow \infty$.
- (iv) $|E_2^*(1 - 1/\tau)| = O(|\tau^2 e^{\pi i \tau}|)$ as $\text{Im}(\tau) \rightarrow \infty$.

Moreover $-\pi^2\theta^4$ has the same properties.

The periodicity (i) of E_2^* is immediate from the definition. The proofs of the other properties of E_2^* are a little more involved.

Consider the forbidden Eisenstein series F and its reverse $\tilde{F}$, which is obtained from reversing the order of summation:

$$F(\tau) = \sum_m \sum_n \frac{1}{(m\tau + n)^2} \quad \text{and} \quad \tilde{F}(\tau) = \sum_n \sum_m \frac{1}{(m\tau + n)^2}.$$

In both cases, the term $n = m = 0$ is omitted.

Lemma 3.9 *The functions F and $\tilde{F}$ satisfy:*

- (a) $F(-1/\tau) = \tau^2 \tilde{F}(\tau)$,
- (b) $F(\tau) - \tilde{F}(\tau) = 2\pi i/\tau$,
- (c) $F(-1/\tau) = \tau^2 F(\tau) - 2\pi i\tau$.

Proof. Property (a) follows directly from the definitions of F and $\tilde{F}$, and the identity

$$(n + m(-1/\tau))^2 = \tau^{-2}(-m + n\tau)^2.$$

To prove property (b), we invoke the functional equation for the Dedekind eta function which was established earlier:

$$\eta(-1/\tau) = \sqrt{\tau/i} \eta(\tau),$$

where $\eta(\tau) = q^{1/12} \prod_{n=1}^{\infty} (1 - q^{2n})$, and $q = e^{\pi i \tau}$.

First, we take the logarithmic derivative of η with respect to the variable τ to find (by Proposition 3.2 in Chapter 5)

$$(\eta'/\eta)(\tau) = \frac{\pi i}{12} - 2\pi i \sum_{n=1}^{\infty} \frac{nq^{2n}}{1 - q^{2n}}.$$

However, if $\sigma_1(k)$ denotes the sum of the divisors of k , then one sees that

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{nq^{2n}}{1 - q^{2n}} &= \sum_{n=1}^{\infty} \sum_{\ell=0}^{\infty} nq^{2n} q^{2\ell n} \\ &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} nq^{2nm} \\ &= \sum_{k=1}^{\infty} \sigma_1(k) q^{2k}. \end{aligned}$$

If we recall that $F(\tau) = \pi^2/3 - 8\pi^2 \sum_{k=1}^{\infty} \sigma_1(k) q^{2k}$, we find

$$(\eta'/\eta)(\tau) = \frac{i}{4\pi} F(\tau).$$

By the chain rule, the logarithmic derivative of $\eta(-1/\tau)$ is $\tau^{-2}(\eta'/\eta)(-1/\tau)$, and using property (a), we see that the logarithmic derivative of $\eta(-1/\tau)$

equals $(i/4\pi)\tilde{F}(\tau)$. Therefore, taking the logarithmic derivative of the functional equation for η we find

$$\frac{i}{4\pi}\tilde{F}(\tau) = \frac{1}{2\tau} + \frac{i}{4\pi}F(\tau),$$

and this gives $\tilde{F}(\tau) = -2\pi i/\tau + F(\tau)$, as desired.

Finally, (c) is a consequence of (a) and (b).

To prove the transformation formula (ii) for E_2^* under $\tau \mapsto -1/\tau$, we begin with

$$E_2^*(\tau) = F(\tau/2) - 4F(2\tau).$$

Then

$$\begin{aligned} E_2^*(-1/\tau) &= F(-1/(2\tau)) - 4F(-2/\tau) \\ &= [4\tau^2 F(2\tau) - 4\pi i\tau] - 4[(\tau/2)^2 F(\tau/2) - \pi i\tau] \\ &= 4\tau^2 F(2\tau) - 4(\tau^2/4)F(\tau/2) \\ &= -\tau^2(F(\tau/2) - 4F(2\tau)) \\ &= -\tau^2 E_2^*(\tau), \end{aligned}$$

as desired. To prove the third property recall that

$$F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum_{k=1}^{\infty} \sigma_1(k) e^{2\pi i k \tau},$$

where the sum goes to 0 as $\text{Im}(\tau) \rightarrow \infty$. Then, if we use the fact that

$$E_2^*(\tau) = F(\tau/2) - 4F(2\tau),$$

we conclude that $E_2^*(\tau) \rightarrow -\pi^2$ as $\text{Im}(\tau) \rightarrow \infty$.

To prove the final property, we begin by showing that

$$(11) \quad E_2^*(1 - 1/\tau) = \tau^2 \left[F\left(\frac{\tau-1}{2}\right) - F(\tau/2) \right].$$

From the transformation formulas for F we have

$$\begin{aligned} F(1/2 - 1/2\tau) &= F\left(\frac{\tau-1}{2\tau}\right) \\ &= \left(\frac{2\tau}{\tau-1}\right)^2 F\left(\frac{2\tau}{1-\tau}\right) - 2\pi i \frac{2\tau}{1-\tau}, \end{aligned}$$

and

$$\begin{aligned} F\left(\frac{2\tau}{1-\tau}\right) &= F(-2 + 2/(1-\tau)) \\ &= F(2/(1-\tau)) \\ &= \left(\frac{1-\tau}{2}\right)^2 F\left(\frac{\tau-1}{2}\right) - 2\pi i \left(\frac{\tau-1}{2}\right). \end{aligned}$$

Hence,

$$F(1/2 - 1/2\tau) = \tau^2 F\left(\frac{\tau-1}{2}\right) - \frac{2\pi i 2\tau}{1-\tau} - 2\pi i \frac{(2\tau)^2}{(\tau-1)^2} \left(\frac{\tau-1}{2}\right).$$

But $F(2 - 2/\tau) = F(-2/\tau) = (\tau^2/4)F(\tau/2) - 2\pi i \tau/2$, and hence

$$\begin{aligned} E_2^*(1 - 1/\tau) &= F(1/2 - 1/2\tau) - 4F(2 - 2/\tau) \\ &= \tau^2 \left[F\left(\frac{\tau-1}{2}\right) - F(\tau/2) \right] - 2\pi i \left(\frac{2\tau}{1-\tau} + \frac{2\tau^2}{\tau-1} \right) + 4\pi i \tau \\ &= \tau^2 \left[F\left(\frac{\tau-1}{2}\right) - F(\tau/2) \right]. \end{aligned}$$

This proves (11). Then, the last property follows from it and the fact that

$$F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum_{k=1}^{\infty} \sigma_1(k) e^{2\pi i k \tau}.$$

Thus Proposition 3.8 is proved.

We can now conclude the proof of the four-squares theorem by considering the quotient $f(\tau) = E_2^*(\tau)/\theta(\tau)^4$, and applying Theorem 3.4, as in the two-squares theorem. Recall $\theta(\tau)^4 \rightarrow 1$ and $\theta(1 - 1/\tau)^4 \sim 16\tau^2 e^{\pi i \tau}$, as $\text{Im}(\tau) \rightarrow \infty$. The result is that $f(\tau)$ is a constant, which equals $-\pi^2$ by Proposition 3.8. This completes the proof of the four-squares theorem.

4 Exercises

1. Prove that

$$\frac{(\Theta'(z|\tau))^2 - \Theta(z|\tau)\Theta''(z|\tau)}{\Theta(z|\tau)^2} = \wp_\tau(z - 1/2 - \tau/2) + c_\tau,$$

where c_τ can be expressed in terms of the first two derivatives of $\Theta(z|\tau)$, with respect to z , at $z = 1/2 + \tau/2$. Compare this formula with the result in Exercise 5 in the previous chapter.

2. Consider the **Fibonacci numbers** $\{F_n\}_{n=0}^\infty$, defined by the two initial values $F_0 = 0$, $F_1 = 1$ and the recursion relation

$$F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 2.$$

- (a) Consider the generating function $F(x) = \sum_{n=0}^\infty F_n x^n$ associated to $\{F_n\}$, and prove that

$$F(x) = x^2 F(x) + xF(x) + x$$

for all x in a neighborhood of 0.

- (b) Show that the polynomial $q(x) = 1 - x - x^2$ can be factored as

$$q(x) = (1 - \alpha x)(1 - \beta x),$$

where α and β are the roots of the polynomial $p(x) = x^2 - x - 1$.

- (c) Expand the expression for F in partial fractions and obtain

$$F(x) = \frac{x}{1 - x - x^2} = \frac{x}{(1 - \alpha x)(1 - \beta x)} = \frac{A}{1 - \alpha x} + \frac{B}{1 - \beta x},$$

where $A = 1/(\alpha - \beta)$ and $B = 1/(\beta - \alpha)$.

- (d) Conclude that $F_n = A\alpha^n + B\beta^n$ for $n \geq 0$. The two roots of p are actually

$$\alpha = \frac{1 + \sqrt{5}}{2} \quad \text{and} \quad \beta = \frac{1 - \sqrt{5}}{2},$$

so that $A = 1/\sqrt{5}$ and $B = -1/\sqrt{5}$.

The number $1/\alpha = (\sqrt{5} - 1)/2$, which is known as the **golden mean**, satisfies the following property: given a line segment $[AC]$ of unit length (Figure 2), there exists a unique point B on this segment so that the following proportion holds

$$\frac{AC}{AB} = \frac{AB}{BC}.$$

If $\ell = AB$, this reduces to the equation $\ell^2 + \ell - 1 = 0$, whose only positive solution is the golden mean. This ratio arises also in the construction of the regular pentagon. It has played a role in architecture and art, going back to the time of ancient Greece.

3. More generally, consider the difference equation given by the initial values u_0 and u_1 , and the recurrence relation $u_n = au_{n-1} + bu_{n-2}$ for $n \geq 2$. Define the generating function associated to $\{u_n\}_{n=0}^\infty$ by $U(x) = \sum_{n=0}^\infty u_n x^n$. The recurrence relation implies that $U(x)(1 - ax - bx^2) = u_0 + (u_1 - au_0)x$ in a neighborhood of

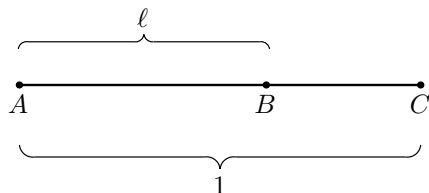


Figure 2. Appearance of the golden mean

the origin. If α and β denote the roots of the polynomial $p(x) = x^2 - ax - b$, then we may write

$$U(x) = \frac{u_0 + (u_1 - au_0)x}{(1 - \alpha x)(1 - \beta x)} = \frac{A}{1 - \alpha x} + \frac{B}{1 - \beta x} = A \sum_{n=0}^{\infty} \alpha^n x^n + B \sum_{n=0}^{\infty} \beta^n x^n,$$

where it is an easy matter to solve for A and B . Finally, this gives $u_n = A\alpha^n + B\beta^n$. Note that this approach yields a solution to our problem if the roots of p are distinct, namely $\alpha \neq \beta$. A variant of the formula holds if $\alpha = \beta$.

4. Using the generating formula for $p(n)$, prove the recurrence formula

$$\begin{aligned} p(n) &= p(n-1) + p(n-2) - p(n-5) - p(n-7) - \cdots \\ &= \sum_{k \neq 0} (-1)^{k+1} p\left(n - \frac{k(3k+1)}{2}\right), \end{aligned}$$

where the right-hand side is the finite sum taken over those $k \in \mathbb{Z}$, $k \neq 0$, with $k(3k+1)/2 \leq n$. Use this formula to calculate $p(5)$, $p(6)$, $p(7)$, $p(8)$, $p(9)$, and $p(10)$; check that $p(10) = 42$.

The next two exercises give elementary results related to the asymptotics of the partition function. More refined statements can be found in Appendix A.

5. Let

$$F(x) = \sum_{n=0}^{\infty} p(n)x^n = \prod_{n=1}^{\infty} \frac{1}{1 - x^n}$$

be the generating function for the partitions. Show that

$$\log F(x) \sim \frac{\pi^2}{6(1-x)} \quad \text{as } x \rightarrow 1, \text{ with } 0 < x < 1.$$

[Hint: Use $\log F(x) = \sum \log(1/(1 - x^n))$ and $\log(1/(1 - x^n)) = \sum (1/m)x^{nm}$, so

$$\log F(x) = \sum \frac{1}{m} \frac{x^m}{1 - x^m}.$$

Use also $mx^{m-1}(1-x) < 1 - x^m < m(1-x).$]

6. Show as a consequence of Exercise 5 that

$$e^{c_1 n^{1/2}} \leq p(n) \leq e^{c_2 n^{1/2}}$$

for two positive constants c_1 and c_2 .

[Hint: $F(e^{-y}) = \sum p(n)e^{-ny} \leq Ce^{c/y}$ as $y \rightarrow 0$. So $p(n)e^{-ny} \leq ce^{c/y}$. Take $y = 1/n^{1/2}$ to get $p(n) \leq c'e^{c'n^{1/2}}$. In the opposite direction

$$\sum_{n=0}^m p(n)e^{-ny} \geq C(e^{c/y} - \sum_{n=m+1}^{\infty} e^{cn^{1/2}}e^{-ny}),$$

and it suffices to take $y = Am^{-1/2}$ where A is a large constant, and use the fact that the sequence $p(n)$ is increasing.]

7. Use the product formula for Θ to prove:

(a) The “triangular number” identity

$$\prod_{n=0}^{\infty} (1+x^n)(1-x^{2n+2}) = \sum_{n=-\infty}^{\infty} x^{n(n+1)/2},$$

which holds for $|x| < 1$.

(b) The “septagonal number” identity

$$\prod_{n=0}^{\infty} (1-x^{5n+1})(1-x^{5n+4})(1-x^{5n+5}) = \sum_{n=-\infty}^{\infty} (-1)^n x^{n(5n+3)/2},$$

which holds for $|x| < 1$.

8. Consider Pythagorean triples (a, b, c) with $a^2 + b^2 = c^2$, and with $a, b, c \in \mathbb{Z}$. Suppose moreover that a and b have no common factors.

(a) Show that either a or b must be odd, and the other even.

(b) Show in this case (assuming a is odd and b even) that there are integers m, n so that $a = m^2 - n^2$, $b = 2mn$, and $c = m^2 + n^2$. [Hint: Note that $b^2 = (c-a)(c+a)$, and prove that $(c-a)/2$ and $(c+a)/2$ are relatively prime integers.]

(c) Conversely, show that whenever c is a sum of two-squares, then there exist integers a and b such that $a^2 + b^2 = c^2$.

9. Use the formula for $r_2(n)$ to prove the following:

(a) If $n = p$, where p is a prime of the form $4k+1$, then $r_2(n) = 8$. This implies that n can be written in a unique way as $n = n_1^2 + n_2^2$, except for the signs and reordering of n_1 and n_2 .

(b) If $n = q^a$, where q is prime of the form $4k+3$ and a is a positive integer, then $r_2(n) > 0$ if and only if a is even.

- (c) In general, n can be represented as the sum of two squares if and only if all the primes of the form $4k+3$ that arise in the prime decomposition of n occur with even exponents.

10. Observe the following irregularities of the functions $r_2(n)$ and $r_4(n)$ as n becomes large:

- (a) $r_2(n) = 0$ for infinitely many n , while $\limsup_{n \rightarrow \infty} r_2(n) = \infty$.
 (b) $r_4(n) = 24$ for infinitely many n while $\limsup_{n \rightarrow \infty} r_4(n)/n = \infty$.

[Hint: For (a) consider $n = 5^k$; for (b) consider alternatively $n = 2^k$, and $n = q^k$ where q is odd and large.]

11. Recall from Problem 2 in Chapter 2, that

$$\sum_{n=1}^{\infty} d(n)z^n = \sum_{n=1}^{\infty} \frac{z^n}{1-z^n}, \quad |z| < 1$$

where $d(n)$ denotes the number of divisors of n .

More generally, show that

$$\sum_{n=1}^{\infty} \sigma_{\ell}(n)z^n = \sum_{n=1}^{\infty} \frac{n^{\ell} z^n}{1-z^n}, \quad |z| < 1$$

where $\sigma_{\ell}(n)$ is the sum of the ℓ^{th} powers of divisors of n .

12. Here we give another identity involving θ^4 , which is equivalent to the four-squares theorem.

- (a) Show that for $|q| < 1$

$$\sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} = \sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)^2}.$$

[Hint: The left-hand side is $\sum \sigma_1(n)q^n$. Use $x/(1-x)^2 = \sum_{n=1}^{\infty} nx^n$.]

- (b) Show as a result that

$$\sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} - \sum_{n=1}^{\infty} \frac{4nq^{4n}}{1-q^{4n}} = \sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)^2} - 4 \sum_{n=1}^{\infty} \frac{q^{4n}}{(1-q^{4n})^2} = \sum \sigma_1^*(n)q^n$$

where $\sigma_1^*(n)$ is the sum of the divisors of d that are not divisible by 4.

- (c) Show that the four-squares theorem is equivalent to the identity

$$\theta(\tau)^4 = 1 + 8 \sum_{n=1}^{\infty} \frac{q^n}{(1 + (-1)^n q^n)^2}, \quad q = e^{\pi i \tau}.$$

5 Problems

1.* Suppose n is of the form $n = 4^a(8k + 7)$, where a and k are positive integers. Show that n cannot be written as the sum of three-squares. The converse, that every n that is *not* of that form *can* be written as the sum of three-squares, is a difficult theorem of Legendre and Gauss.

2. Let $\mathrm{SL}_2(\mathbb{Z})$ denote the set of 2×2 matrices with integer entries and determinant 1, that is,

$$\mathrm{SL}_2(\mathbb{Z}) = \left\{ g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z} \text{ and } ad - bc = 1 \right\}.$$

This group acts on the upper half-plane by the fractional linear transformation $g(\tau) = (a\tau + b)/(c\tau + d)$. Together with this action comes the so-called fundamental domain $\mathcal{F}_1$ in the complex plane defined by

$$\mathcal{F}_1 = \{\tau \in \mathbb{C} : |\tau| \geq 1, \quad |\mathrm{Re}(\tau)| \leq 1/2 \text{ and } |\mathrm{Im}(\tau)| \geq 0\}.$$

It is illustrated in Figure 3.

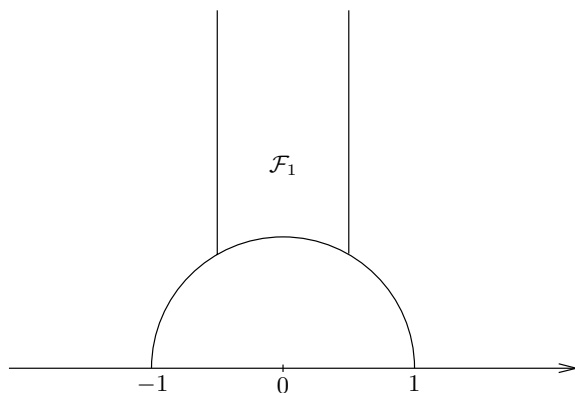


Figure 3. The fundamental domain $\mathcal{F}_1$

Consider the two elements in $\mathrm{SL}_2(\mathbb{Z})$ defined by $S(\tau) = -1/\tau$ and $T_1(\tau) = \tau + 1$. These correspond (for example) to the matrices

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

respectively. Let $\mathfrak{g}$ be the subgroup of $\mathrm{SL}_2(\mathbb{Z})$ generated by S and T_1 .

- Show that for every $\tau \in \mathbb{H}$ there exists $g \in \mathfrak{g}$ such that $g(\tau) \in \mathcal{F}_1$.
- We say that two points τ and τ' are congruent if there exists $g \in \mathrm{SL}_2(\mathbb{Z})$ such that $g(\tau) = \tau'$. Prove that if $\tau, w \in \mathcal{F}_1$ are congruent, then either $\mathrm{Re}(\tau) =$

$\pm 1/2$ and $\tau' = \tau \mp 1$ or $|\tau| = 1$ and $\tau' = -1/\tau$. [Hint: Say $\tau' = g(\tau)$. Why can one assume that $\text{Im}(\tau') \geq \text{Im}(\tau)$, and therefore $|c\tau + d| \leq 1$? Now consider separately the possibilities $c = -1$, $c = 0$, or $c = 1$.]

- (c) Prove that S and T_1 generate the modular group in the sense that every fractional linear transformation corresponding to $g \in \text{SL}_2(\mathbb{Z})$ is a composition of finitely many S 's and T_1 's, and their inverses. Strictly speaking, the matrices associated to S and T_1 generate the projective special linear group $\text{PSL}_2(\mathbb{Z})$, which equals $\text{SL}_2(\mathbb{Z})$ modulo $\pm I$. [Hint: Observe that $2i$ is in the interior of $\mathcal{F}_1$. Now map $g(2i)$ back into $\mathcal{F}_1$ by using part (a). Use part (b) to conclude.]

3. In this problem, consider the group G of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with integer entries, determinant 1, and such that a and d have the same parity, b and c have the same parity, and c and d have opposite parity. This group also acts on the upper half-plane by fractional linear transformations. To the group G corresponds the fundamental domain $\mathcal{F}$ defined by $|\tau| \geq 1$, $|\text{Re}(\tau)| \leq 1$, and $\text{Im}(\tau) \geq 0$ (see Figure 1). Also, let

$$S(\tau) = -1/\tau \leftrightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad T_2(\tau) = \tau + 2 \leftrightarrow \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Prove that every fractional linear transformation corresponding to $g \in G$ is a composition of finitely many S , T_2 and their inverses, in analogy with the previous problem.

4. Let G denote the group of matrices given in the previous problem. Here we give an alternate proof of Theorem 3.4, that states that a function in $\mathbb{H}$ which is holomorphic, bounded, and invariant under G must be constant.

- (a) Suppose that $f: \mathbb{H} \rightarrow \mathbb{C}$ is holomorphic, bounded, and that there exists a sequence of complex numbers $\tau_k = x_k + iy_k$ such that

$$f(\tau_k) = 0, \quad \sum_{k=1}^{\infty} y_k = \infty, \quad 0 < y_k \leq 1, \quad \text{and} \quad |x_k| \leq 1.$$

Then $f = 0$. [Hint: When $x_k = 0$ see Problem 5 in Chapter 8.]

- (b) Given two relatively prime integers c and d with different parity, show that there exist integers a and b such that $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$. [Hint: All the solutions of $xc + dy = 1$ take the form $x_0 + dt$ and $y_0 - ct$ where x_0, y_0 is a particular solution and $t \in \mathbb{Z}$.]
- (c) Prove that $\sum 1/(c^2 + d^2) = \infty$ where the sum is taken over all c and d that are relatively prime and of opposite parity. [Hint: Suppose not, and prove that $\sum_{(a,b)=1} 1/(a^2 + b^2) < \infty$ where the sum is over all relatively prime integers a and b . To do so, note that if a and b are both odd and relatively prime, then the two numbers c and d defined by $c = (a+b)/2$

and $d = (a - b)/2$ are relatively prime and of opposite parity. Moreover, $c^2 + d^2 \leq A(a^2 + b^2)$ for some universal constant A . Therefore

$$\sum_{n \neq 0} \frac{1}{n^2} \sum_{(a,b)=1} \frac{1}{a^2 + b^2} < \infty,$$

hence $\sum 1/(k^2 + \ell^2) < \infty$, where the sum is over all integers k and ℓ such that $k, \ell \neq 0$. Why is this a contradiction?]

- (d) Prove that if $F: \mathbb{H} \rightarrow \mathbb{C}$ is holomorphic, bounded, and invariant under G , then F is constant. [Hint: Replace $F(\tau)$ by $F(\tau) - F(i)$ so that we can assume $F(i) = 0$ and prove $F = 0$. For each relatively prime c and d with opposite parity, choose $g \in G$ so that $g(i) = x_{c,d} + i/(c^2 + d^2)$ with $|x_{c,d}| \leq 1$.]

5.* In Chapter 9 we proved that the Weierstrass $\wp$ function satisfies the cubic equation

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3,$$

where $g_2 = 60E_4$, $g_3 = 140E_6$, with E_k is the Eisenstein series of order k . The discriminant of the cubic $y^2 = 4x^3 - g_2x - g_3$ is defined by $\Delta = g_2^3 - 27g_3^2$. Prove that

$$\Delta(\tau) = (2\pi)^{12} \eta^{24}(\tau) \quad \text{for all } \tau \in \mathbb{H}.$$

[Hint: Δ and η^{24} satisfy the same transformation laws under $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$. Because of the fundamental domain described in Problem 2, it suffices then to investigate the behavior at the only cusp, which is at infinity.]

6.* Here we will deduce the formula for $r_8(n)$, which counts the number of representations of n as a sum of eight squares. The method is parallel to that of $r_4(n)$, but the details are less delicate.

Theorem: $r_8(n) = 16\sigma_3^*(n)$.

Here $\sigma_3^*(n) = \sigma_3(n) = \sum_{d|n} d^3$, when n is odd. Also, when n is even

$$\sigma_3^*(n) = \sum_{d|n} (-1)^d d^3 = \sigma_3^e(n) - \sigma_3^o(n),$$

where $\sigma_3^e(n) = \sum_{d|n, d \text{ even}} d^3$ and $\sigma_3^o(n) = \sum_{d|n, d \text{ odd}} d^3$.

Consider the appropriate Eisenstein series

$$E_4^*(\tau) = \sum \frac{1}{(n + m\tau)^4},$$

where the sum is over integers n and m with opposite parity. Recall the standard Eisenstein series

$$E_4(\tau) = \sum_{(n,m) \neq (0,0)} \frac{1}{(n + m\tau)^4}.$$

Notice that the series defining E_4^* is absolutely convergent, in distinction to $E_2^*(\tau)$, which arose when considering $r_4(n)$. This makes some of the considerations below quite a bit simpler.

- (a) Prove that $r_8(n) = 16\sigma_3^*(n)$ is equivalent to the identity $\theta(\tau)^8 = 48\pi^{-4}E_4^*(\tau)$.
 [Hint: Use the fact that $E_4(\tau) = 2\zeta(4) + \frac{(2\pi)^4}{3} \sum_{k=1}^{\infty} \sigma_3(k)e^{2\pi i k \tau}$ and $\zeta(4) = \pi^4/90$.]
- (b) Note that $E_4^*(\tau) = E_4(\tau) - 2^{-4}E_4((\tau - 1)/2)$.
- (c) $E_4^*(\tau + 2) = E_4^*(\tau)$.
- (d) $E_4^*(\tau) = \tau^{-4}E_4^*(-1/\tau)$.
- (e) $(48/\pi^4)E_4^*(\tau) \rightarrow 1$ as $\tau \rightarrow \infty$.
- (f) $|E_4^*(1 - 1/\tau)| \approx |\tau|^4|e^{2\pi i \tau}|$, as $\text{Im}(\tau) \rightarrow \infty$. [Hint: Verify that $E_4^*(1 - 1/\tau) = \tau^4(E_4(\tau) - E_4(2\tau))$.]

Since $\theta(\tau)^8$ satisfies properties similar to (c), (d), (e) and (f) above, it follows that the invariant function $48\pi^{-4}E_4^*(\tau)/\theta(\tau)^8$ is bounded and hence a constant, which must be 1. This gives the desired result.

Appendix A: Asymptotics

On the numerical computation of the definite integral $\int_w \cos \frac{\pi}{2}(w^3 - m.w)$, between the limits 0 and $\frac{1}{0}$.

The simplicity of the form of this differential coefficient induces me to suppose that the integral may possibly be expressible by some of the integrals whose values have been tabulated. After many attempts however, I have not succeeded in reducing it to any known integral: and I have therefore computed its value by actual summation to a considerable extent and by series for the remainder.

G. B. Airy, 1838

In a number of problems in analysis the solution is given by a function whose explicit calculation is not tractable. Often a useful substitute (and the only recourse) is to study the asymptotic behavior of this function near the point of interest. Here we shall investigate several related types of asymptotics, where the ideas of complex analysis are of crucial help. These typically center about the behavior for large values of the variable s of an integral of the form

$$(1) \qquad I(s) = \int_a^b e^{-s\Phi(x)} dx.$$

We organize our presentation by formulating three guiding principles.

- (i) **Deformation of contour.** The function Φ is in general complex-valued, therefore, for large s the integrand in (1) may oscillate rapidly, so that the resulting cancellations mask the true behavior of $I(s)$. When Φ is holomorphic (which is often the case) one can hope to change the contour of integration so that as far as possible, on the new contour Φ is essentially real-valued. If this is possible, one can then hope to read off the behavior of $I(s)$ in a rather direct manner. This idea will be illustrated first in the context of Bessel functions.
- (ii) **Laplace's method.** In the case when Φ is real-valued on the contour and s is positive, the maximum contribution to $I(s)$ comes

from the integration near a minimum of Φ , and this leads to a satisfactory expansion in terms of the quadratic behavior of Φ near its minimum. We apply these ideas to present the asymptotics of the gamma function (Stirling's formula), and also those of the Airy function.

- (iii) **Generating functions.** If $\{F_n\}$ is a number-theoretic or combinatorial sequence, we have already seen in several examples that one can exploit analytic properties of the generating function, $F(u) = \sum F_n u^n$, to obtain interesting conclusions regarding $\{F_n\}$. In fact the asymptotic behavior of F_n , as $n \rightarrow \infty$, can also be analyzed this way, via the formula

$$F_n = \int_{\gamma} F(e^{2\pi iz}) e^{-2\pi inz} dz.$$

Here γ is an appropriate segment of unit length in the upper half-plane. This formula can then be studied as a variant of the integral (1). We shall show how these ideas apply in an important particular case to obtain an asymptotic formula for $p(n)$, the number of partitions of n .

1 Bessel functions

Bessel functions appear naturally in many problems that exhibit rotational symmetries. For instance, the Fourier transform of a spherical function in $\mathbb{R}^d$ is neatly expressed in terms of a Bessel function of order $(d/2) - 1$. See Chapter 6 in Book I.

The Bessel functions can be defined by a number of alternative formulas. We take the one that is valid for all order $\nu > -1/2$, given by

$$(2) \quad J_{\nu}(s) = \frac{(s/2)^{\nu}}{\Gamma(\nu + 1/2)\Gamma(1/2)} \int_{-1}^1 e^{isx} (1 - x^2)^{\nu-1/2} dx.$$

If we also write $J_{-1/2}(s)$ for $\lim_{\nu \rightarrow -1/2} J_{\nu}(s)$, we see that it equals $\sqrt{\frac{2}{\pi s}} \cos s$; observe in addition that $J_{1/2}(s) = \sqrt{\frac{2}{\pi s}} \sin s$. However, $J_{\nu}(s)$ has an expression in terms of elementary functions only when ν is half-integral, and understanding this function in general requires further analysis. Its behavior for large s is suggested by the two examples above.

Theorem 1.1 $J_{\nu}(s) = \sqrt{\frac{2}{\pi s}} \cos\left(s - \frac{\pi\nu}{2} - \frac{\pi}{4}\right) + O(s^{-3/2})$ as $s \rightarrow \infty$.

In view of the formula for $J_\nu(s)$ it suffices to investigate

$$(3) \quad I(s) = \int_{-1}^1 e^{isx} (1-x^2)^{\nu-1/2} dx,$$

and to this end we consider the analytic function $f(z) = e^{isz}(1-z^2)^{\nu-1/2}$ in the complex plane slit along the rays $(-\infty, -1) \cup (1, \infty)$; for $(1-z^2)^{\nu-1/2}$ we choose that branch that is positive when $z = x \in (-1, 1)$. With $s > 0$ fixed, we apply Cauchy's theorem to see that

$$I(s) = -I_-(s) - I_+(s),$$

where the integrals $I(s)$, $I_-(s)$, and $I_+(s)$ are taken over the lines shown in Figure 1. This is established by using the fact that $\int_{\gamma_{\epsilon,R}} f(z) dz = 0$ where $\gamma_{\epsilon,R}$ is the second contour of Figure 1, and letting $\epsilon \rightarrow 0$ and $R \rightarrow \infty$.

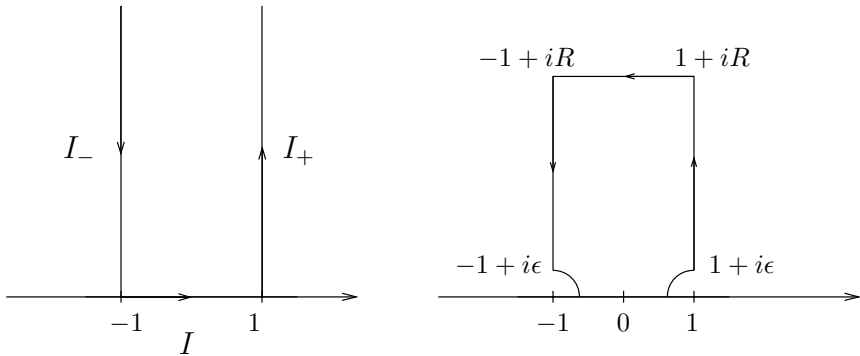


Figure 1. Contours of integration of $I(s)$, $I_-(s)$, $I_+(s)$, and the contour $\gamma_{\epsilon,R}$

On the contour for $I_+(s)$ we have $z = 1 + iy$, so

$$(4) \quad I_+(s) = ie^{is} \int_0^\infty e^{-sy} (1 - (1+iy)^2)^{\nu-1/2} dy.$$

There is a similar expression for $I_-(s)$.

What has the passage from $I(s)$ to $-(I_-(s) + I_+(s))$ gained us? Observe that for large positive s , the exponential e^{isx} in (3) oscillates rapidly, so the estimation of that integral is not obvious at first glance. However, in (4) the corresponding exponential is e^{-sy} , and it decreases

rapidly as $s \rightarrow \infty$, except when $y = 0$. Thus in this case one sees immediately that the main contribution to the integral comes from the integration near $y = 0$, and this allows one readily to approximate this integral. This idea is made precise in the following observation.

Proposition 1.2 *Suppose a and m are fixed, with $a > 0$ and $m > -1$. Then as $s \rightarrow \infty$*

$$(5) \quad \int_0^a e^{-sx} x^m dx = s^{-m-1} \Gamma(m+1) + O(e^{-cs}),$$

for some positive c .

Proof. The fact that $m > -1$ guarantees that $\int_0^a e^{-sx} x^m dx = \lim_{\epsilon \rightarrow 0} \int_\epsilon^a e^{-sx} x^m dx$ exists. Then, we write

$$\int_0^a e^{-sx} x^m dx = \int_0^\infty e^{-sx} x^m dx - \int_a^\infty e^{-sx} x^m dx.$$

The first integral on the right-hand side can be seen to equal $s^{-m-1} \Gamma(m+1)$, if we make the change of variables $x \mapsto x/s$. For the second integral we note that

$$(6) \quad \int_a^\infty e^{-sx} x^m dx = e^{-cs} \int_a^\infty e^{-s(x-c)} x^m dx = O(e^{-cs}),$$

as long as $c < a$, and so the proposition is proved.

We return to the integral (4) and observe that

$$(1 - (1 + iy)^2)^{\nu-1/2} = (-2iy)^{\nu-1/2} + O(y^{\nu+1/2}), \quad \text{for } 0 \leq y \leq 1,$$

while

$$(1 - (1 + iy)^2)^{\nu-1/2} = O(y^{\nu-1/2} + y^{2\nu-1}), \quad \text{for } 1 \leq y.$$

So, applying the proposition with $a = 1$ and $m = \nu \mp 1/2$, as well as (6), gives

$$I_+(s) = i(-2i)^{\nu-1/2} e^{is} s^{-\nu-1/2} \Gamma(\nu + 1/2) + O(s^{-\nu-3/2}).$$

Similarly,

$$I_-(s) = i(2i)^{\nu-1/2} e^{is} s^{-\nu-1/2} \Gamma(\nu + 1/2) + O(s^{-\nu-3/2}).$$

If we recall that

$$J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu + 1/2)\Gamma(1/2)}[-I_-(s) - I_+(s)],$$

and the fact that $\Gamma(1/2) = \sqrt{\pi}$, we see that we have obtained the proof of the theorem.

For later purposes it is interesting to point out that under certain restricted circumstances, the gist of the conclusion in Proposition 1.2 extends to the complex half-plane $\operatorname{Re}(s) \geq 0$.

Proposition 1.3 *Suppose a and m are fixed, with $a > 0$ and $-1 < m < 0$. Then as $|s| \rightarrow \infty$ with $\operatorname{Re}(s) \geq 0$,*

$$\int_0^a e^{-sx} x^m dx = s^{-m-1} \Gamma(m+1) + O(1/|s|).$$

(Here s^{-m-1} is the branch of that function that is positive for $s > 0$).

Proof. We begin by showing that when $\operatorname{Re}(s) \geq 0$, $s \neq 0$,

$$\int_0^\infty e^{-sx} x^m dx = \lim_{N \rightarrow \infty} \int_0^N e^{-sx} x^m dx$$

exists and equals $s^{-m-1} \Gamma(m+1)$. If N is large, we first write

$$\int_0^N e^{-sx} x^m dx = \int_0^a e^{-sx} x^m dx + \int_a^N e^{-sx} x^m dx.$$

Since $m > -1$, the first integral on the right-hand side defines an analytic function everywhere. For the second integral, we note that $-\frac{1}{s} \frac{d}{dx}(e^{-sx}) = e^{-sx}$, so an integration by parts gives

$$(7) \quad \int_a^N e^{-sx} x^m dx = \frac{m}{s} \int_a^N e^{-sx} x^{m-1} dx - \left[\frac{e^{-sx}}{s} x^m \right]_a^N.$$

This identity, together with the convergence of the integral $\int_a^\infty x^{m-1} dx$, shows that $\int_a^\infty e^{-sx} x^m dx$ defines an analytic function on $\operatorname{Re}(s) > 0$ that is continuous on $\operatorname{Re}(s) \geq 0$, $s \neq 0$. Thus $\int_0^\infty e^{-sx} x^m dx$ is analytic on the half-plane $\operatorname{Re}(s) > 0$ and continuous on $\operatorname{Re}(s) \geq 0$, $s \neq 0$. Since it equals $s^{-m-1} \Gamma(m+1)$ when s is positive, we deduce that $\int_0^\infty e^{-sx} x^m dx = s^{-m-1} \Gamma(m+1)$ when $\operatorname{Re}(s) \geq 0$, $s \neq 0$.

However, we now have

$$\int_0^a e^{-sx} x^m dx = \int_0^\infty e^{-sx} x^m dx - \int_a^\infty e^{-sx} x^m dx.$$

It is clear from (7), and from the fact that $m < 0$, that if we let $N \rightarrow \infty$, then $\int_a^\infty e^{-sx} x^{m-1} dx = O(1/|s|)$. The proposition is therefore proved.

Note. If one wants to obtain a better error term in Proposition 1.3, or for that matter extend the range of m , then one needs to mitigate the effect of the contribution of the end-point $x = a$. This can be done by introducing suitable smooth cut-offs. See Problem 1.

2 Laplace's method; Stirling's formula

We have already mentioned that when Φ is real-valued, the main contribution to $\int_a^b e^{-s\Phi(x)} dx$ as $s \rightarrow \infty$ comes from the point where Φ takes its minimum value. A situation where this minimum is attained at one of the end-points, a or b , was considered in Proposition 1.2. We now turn to the important case when the minimum is achieved in the interior of $[a, b]$.

Consider

$$\int_a^b e^{-s\Phi(x)} \psi(x) dx$$

where the **phase** Φ is real-valued, and both it and the **amplitude** ψ are assumed for simplicity to be indefinitely differentiable. Our hypothesis regarding the minimum of Φ is that there is an $x_0 \in (a, b)$ so that $\Phi'(x_0) = 0$, but $\Phi''(x_0) > 0$ throughout $[a, b]$ (Figure 2 illustrates the situation.)

Proposition 2.1 *Under the above assumptions, with $s > 0$ and $s \rightarrow \infty$,*

$$(8) \quad \int_a^b e^{-s\Phi(x)} \psi(x) dx = e^{-s\Phi(x_0)} \left[\frac{A}{s^{1/2}} + O\left(\frac{1}{s}\right) \right],$$

where

$$A = \sqrt{2\pi} \frac{\psi(x_0)}{(\Phi''(x_0))^{1/2}}.$$

Proof. By replacing $\Phi(x)$ by $\Phi(x) - \Phi(x_0)$ we may assume that $\Phi(x_0) = 0$. Since $\Phi'(x_0) = 0$, we note that

$$\frac{\Phi(x)}{(x - x_0)^2} = \frac{\Phi''(x_0)}{2} \varphi(x),$$

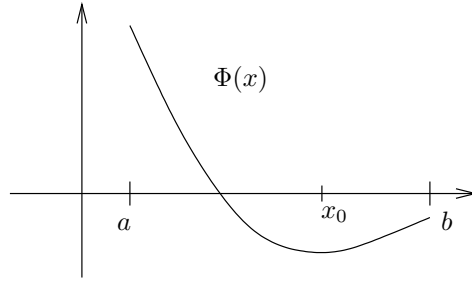


Figure 2. The function Φ , with its minimum at x_0

where φ is smooth, and $\varphi(x) = 1 + O(x - x_0)$ as $x \rightarrow x_0$. We can therefore make the smooth change of variables $x \mapsto y = (x - x_0)(\varphi(x))^{1/2}$ in a small neighborhood of $x = x_0$, and observe that $dy/dx|_{x_0} = 1$, and thus $dx/dy = 1 + O(y)$ as $y \rightarrow 0$. Moreover, we have $\psi(x) = \tilde{\psi}(y)$ with $\tilde{\psi}(y) = \psi(x_0) + O(y)$ as $y \rightarrow 0$. Hence if $[a', b']$ is a sufficiently small interval containing x_0 in its interior, by making the indicated change of variables we obtain

$$(9) \quad \int_{a'}^{b'} e^{-s\Phi(x)} \psi(x) dx = \psi(x_0) \int_{\alpha}^{\beta} e^{-s \frac{\Phi''(x_0)}{2} y^2} dy + O \left(\int_{\alpha}^{\beta} e^{-s \frac{\Phi''(x_0)}{2} y^2} |y| dy \right),$$

where $\alpha < 0 < \beta$. We now make the further change of variables $y^2 = X$, $dy = \frac{1}{2} X^{-1/2} dX$, and we see by (5) that the first integral on the right-hand side in (9) is

$$\int_0^{a_0} e^{-s \frac{\Phi''(x_0)}{2} X} X^{-1/2} dX + O(e^{-\delta s}) = s^{-1/2} \left(\frac{2\pi}{\Phi''(x_0)} \right)^{1/2} + O(e^{-\delta s}),$$

for some $\delta > 0$. By the same argument, the second integral is $O(1/s)$. What remains are the integrals of $e^{-s\Phi(x)} \psi(x)$ over $[a, a']$ and $[b', b]$; but these integrals decay exponentially as $s \rightarrow \infty$, since $\Phi(x) \geq c > 0$ in these two sub-intervals. Altogether, this establishes (8) and the proposition.

It is important to realize that the asymptotic relation (8) extends to all complex s with $\text{Re}(s) \geq 0$. The proof, however, requires a somewhat different argument: here we must take into account the oscillations of $e^{-s\Phi(x)}$ when $|s|$ is large but $\text{Re}(s)$ is small, and this is achieved by a simple integration by parts.

Proposition 2.2 *With the same assumptions on Φ and ψ , the relation (8) continues to hold if $|s| \rightarrow \infty$ with $\operatorname{Re}(s) \geq 0$.*

Proof. We proceed as before to the equation (9), and obtain the appropriate asymptotic for the first term, by virtue of Proposition 1.3, with $m = -1/2$. To deal with the rest we start with an observation. If Ψ and ψ are given on an interval $[\bar{a}, \bar{b}]$, are indefinitely differentiable, and $\Psi(x) \geq 0$, while $|\Psi'(x)| \geq c > 0$, then if $\operatorname{Re}(s) \geq 0$,

$$(10) \quad \int_{\bar{a}}^{\bar{b}} e^{-s\Psi(x)} \psi(x) dx = O\left(\frac{1}{|s|}\right) \quad \text{as } |s| \rightarrow \infty.$$

Indeed, the integral equals

$$-\frac{1}{s} \int_{\bar{a}}^{\bar{b}} \frac{d}{dx} (e^{-s\Psi(x)}) \frac{\psi(x)}{\Psi'(x)} dx,$$

which by integration by parts gives

$$\frac{1}{s} \int_{\bar{a}}^{\bar{b}} e^{-s\Psi(x)} \frac{d}{dx} \left(\frac{\psi(x)}{\Psi'(x)} \right) dx - \frac{1}{s} \left[e^{-s\Psi(x)} \frac{\psi(x)}{\Psi'(x)} \right]_{\bar{a}}^{\bar{b}}.$$

The assertion (10) follows immediately since $|e^{-s\Psi(x)}| \leq 1$, when $\operatorname{Re}(s) \geq 0$. This allows us to deal with the integrals of $e^{-s\Psi(x)} \psi(x)$ in the complementary intervals $[a, a']$ and $[b', b]$, because in each, $|\Phi'(x)| \geq c > 0$, since $\Phi'(x_0) = 0$ and $\Phi''(x) \geq c_1 > 0$.

Finally, for the second term on the right-hand side of (9) we observe that it is actually of the form

$$\int_{\alpha}^{\beta} e^{-s \frac{\Phi''(x_0)}{2} y^2} y \eta(y) dy,$$

where $\eta(y)$ is differentiable. Then we can again estimate this term by integration by parts, once we write it as

$$-\frac{1}{s\Phi''(x_0)} \int_{\alpha}^{\beta} \frac{d}{dy} \left(e^{-s \frac{\Phi''(x_0)}{2} y^2} \right) \eta(y) dy,$$

obtaining the bound $O(1/|s|)$.

The special case of Proposition 2.2 when s is purely imaginary, $s = it$, $t \rightarrow \pm\infty$, is often treated separately; the argument in this situation is

usually referred to as the method of **stationary phase**. The points x_0 for which $\Phi'(x_0) = 0$ are called the **critical points**.

Our first application will be to the asymptotic behavior of the gamma function Γ , given by Stirling's formula. This formula will be valid in any sector of the complex plane that omits the negative real axis. For any $\delta > 0$ we set $S_\delta = \{s : |\arg s| \leq \pi - \delta\}$, and denote by $\log s$ the principal branch of the logarithm that is given in the plane slit along the negative real axis.

Theorem 2.3 *If $|s| \rightarrow \infty$ with $s \in S_\delta$, then*

$$(11) \quad \Gamma(s) = e^{s \log s} e^{-s} \frac{\sqrt{2\pi}}{s^{1/2}} \left(1 + O\left(\frac{1}{|s|^{1/2}}\right) \right).$$

Remark. With a little extra effort one can improve the error term to $O(1/|s|)$, and in fact obtain a complete asymptotic expansion in powers of $1/s$; see Problem 2. Also, we note that (11) implies $\Gamma(s) \sim \sqrt{2\pi} s^{s-1/2} e^{-s}$, which is how Stirling's formula is often stated.

To prove the theorem we first establish (11) in the right half-plane. We shall show that the formula holds whenever $\operatorname{Re}(s) > 0$, and in addition that the error term is uniform on the closed half-plane, once we omit a neighborhood of the origin (say $|s| < 1$). To see this, start with $s > 0$, and write

$$\Gamma(s) = \int_0^\infty e^{-x} x^s \frac{dx}{x} = \int_0^\infty e^{-x+s \log x} \frac{dx}{x}.$$

Upon making the change of variables $x \mapsto sx$, the above equals

$$\int_0^\infty e^{-sx+s \log sx} \frac{dx}{x} = e^{s \log s} e^{-s} \int_0^\infty e^{-s\Phi(x)} \frac{dx}{x},$$

where $\Phi(x) = x - 1 - \log x$. By analytic continuation this identity continues to hold, and we have when $\operatorname{Re}(s) > 0$,

$$\Gamma(s) = e^{s \log s} e^{-s} I(s)$$

with

$$I(s) = \int_0^\infty e^{-s\Phi(x)} \frac{dx}{x}.$$

It now suffices to see that

$$(12) \quad I(s) = \frac{\sqrt{2\pi}}{s^{1/2}} + O\left(\frac{1}{|s|}\right) \quad \text{for } \operatorname{Re}(s) > 0.$$

Observe first that $\Phi(1) = \Phi'(1) = 0$, $\Phi''(x) = 1/x^2 > 0$ whenever $0 < x < \infty$, and $\Phi''(1) = 1$. Thus Φ is convex, attains its minimum at $x = 1$, and is positive.

We apply the complex version of the Laplace method, Proposition 2.2, in this situation. Here the critical point is $x_0 = 1$ and $\psi(x) = 1/x$. We choose for convenience the interval $[a, b]$ to be $[1/2, 2]$. Then for $\int_a^b e^{-s\Phi(x)}\psi(x) dx$ we get the asymptotic (12). It remains to bound the error terms, those corresponding to integration over $[0, 1/2]$, and $[2, \infty)$. Here the device of integration by parts, which has served us so well, can be applied again. Indeed, since $\Phi'(x) = 1 - 1/x$, we have

$$\begin{aligned} \int_{\epsilon}^{1/2} e^{-s\Phi(x)} \frac{dx}{x} &= -\frac{1}{s} \int_{\epsilon}^{1/2} \frac{d}{dx} (e^{-s\Phi(x)}) \frac{dx}{\Phi'(x)x} \\ &= -\frac{1}{s} \left[\frac{e^{-s\Phi(x)}}{x-1} \right]_{\epsilon}^{1/2} - \frac{1}{s} \int_{\epsilon}^{1/2} e^{-s\Phi(x)} \frac{dx}{(x-1)^2}. \end{aligned}$$

Noting that $\Phi(\epsilon) \rightarrow +\infty$ as $\epsilon \rightarrow 0$, and $|e^{-s\Phi(x)}| \leq 1$, we find in the limit that

$$\int_0^{1/2} e^{-s\Phi(x)} \frac{dx}{x} = \frac{2}{s} e^{-s\Phi(1/2)} - \frac{1}{s} \int_0^{1/2} e^{-s\Phi(x)} \frac{dx}{(x-1)^2}.$$

Thus the left-hand side is $O(1/|s|)$ in the half-plane $\operatorname{Re}(s) \geq 0$.

The integral $\int_2^{\infty} e^{-s\Phi(x)} \frac{dx}{x}$ is treated analogously, once we note that $\int_2^{\infty} (x-1)^{-2} dx$ converges.

Since these estimates are uniform, (12) and thus (11) are proved for $\operatorname{Re}(s) \geq 0$, $|s| \rightarrow \infty$.

To pass from $\operatorname{Re}(s) \geq 0$ to $\operatorname{Re}(s) \leq 0$, $s \in S_{\delta}$, we record the following fact about the principal branch of $\log s$: whenever $\operatorname{Re}(s) \geq 0$, $s = \sigma + it$, $t \neq 0$, then

$$\log(-s) = \begin{cases} \log s - i\pi & \text{if } t > 0, \\ \log s + i\pi & \text{if } t < 0. \end{cases}$$

Hence if $G(s) = e^{s \log s} e^{-s}$, $\operatorname{Re}(s) \geq 0$, $t \neq 0$, then

$$(13) \quad G(-s)^{-1} = \begin{cases} e^{s \log s} e^{-s} e^{-si\pi} & \text{if } t > 0, \\ e^{s \log s} e^{-s} e^{si\pi} & \text{if } t < 0. \end{cases}$$

Next,

$$(14) \quad \Gamma(s)\Gamma(-s) = \frac{\pi}{-s \sin \pi s},$$

which follows from the fact that $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$, and $\Gamma(1-s) = -s\Gamma(-s)$ (see Theorem 1.4 and Lemma 1.2 in Chapter 6). The combination of (13) and (14), together with the fact that for large s , $(1 + O(1/|s|^{1/2}))^{-1} = 1 + O(1/|s|^{1/2})$, allows us then to extend (11) to the whole sector S_δ , thereby completing the proof of the theorem.

3 The Airy function

The Airy function appeared first in optics, and more precisely, in the analysis of the intensity of light near a caustic; it was an important early instance in the study of asymptotics of integrals, and it continues to arise in a number of other problems. The **Airy function** Ai is defined by

$$(15) \quad \text{Ai}(s) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(x^3/3+sx)} dx, \quad \text{with } s \in \mathbb{R}.$$

Let us first see that because of the rapid oscillations of the integrand as $|x| \rightarrow \infty$, the integral converges and represents a continuous function of s . In fact, note that

$$\frac{1}{i(x^2 + s)} \frac{d}{dx} \left(e^{i(x^3/3+sx)} \right) = e^{i(x^3/3+sx)},$$

so if $a \geq 2|s|^{1/2}$, we can write the integral $\int_a^R e^{i(x^3/3+sx)} dx$ as

$$(16) \quad \int_a^R \frac{1}{i(x^2 + s)} \frac{d}{dx} \left(e^{i(x^3/3+sx)} \right) dx.$$

We may now integrate by parts and let $R \rightarrow \infty$, to see that the integral converges uniformly, and that as a result $\int_a^\infty e^{i(x^3/3+sx)} dx$ is also continuous for $|s| \leq a^2/4$. The same argument works for the integral from $-\infty$ to $-a$ and our assertion regarding $\text{Ai}(s)$ is established.

A better insight into $\text{Ai}(s)$ is given by deforming the contour of integration in (15). A choice of an optimal contour will appear below, but for now let us notice that as soon as we replace the x -axis of integration in (15) by the parallel line $L_\delta = \{x + i\delta, x \in \mathbb{R}\}$, $\delta > 0$, matters improve dramatically.

In fact, we may apply the Cauchy theorem to $f(z) = e^{i(z^3/3+sz)}$ over the rectangle γ_R shown in Figure 3.

One observes that $f(z) = O(e^{-\delta x^2})$ on L_δ , while $f(z) = O(e^{-yR^2})$ on the vertical sides of the rectangle. Thus since $\int_0^\delta e^{-yR^2} dy \rightarrow 0$ as



Figure 3. The line L_δ and the contour γ_R

$R \rightarrow \infty$, we see that

$$\text{Ai}(s) = \frac{1}{2\pi} \int_{L_\delta} e^{i(z^3/3 + sz)} dz.$$

Now the majorization $f(z) = O(e^{-\delta x^2})$ continues to hold for each complex s , and hence because of the (rapid) convergence of the integral, $\text{Ai}(s)$ extends to an entire function of s .

We note next that $\text{Ai}(s)$ satisfies the differential equation

$$(17) \quad \text{Ai}''(s) = s\text{Ai}(s).$$

This simple and natural equation helps to explain the ubiquity of the Airy function. To prove (17) observe that

$$\text{Ai}''(s) - s\text{Ai}(s) = \frac{1}{2\pi} \int_{L_\delta} (-z^2 - s)e^{i(z^3/3 + sz)} dz.$$

But $-(z^2 + s)e^{i(z^3/3 + sz)} = i \frac{d}{dz}(e^{i(z^3/3 + sz)})$, so

$$\text{Ai}''(s) - s\text{Ai}(s) = \frac{i}{2\pi} \int_{L_\delta} \frac{d}{dz}(f(z)) dz = 0,$$

since $f(z) = e^{i(z^3/3 + sz)}$ vanishes as $|z| \rightarrow \infty$ along L_δ .

We now turn to our main problem, the asymptotics of $\text{Ai}(s)$ for large (real) values of s . The differential equation (17) shows us that we may expect different behaviors of the Airy function when $|s|$ is large, depending on whether s is positive or negative. To see this, we compare the equation with a simple analogue

$$(18) \quad y''(s) = Ay(s),$$

where A is a large constant, with A positive when considering s positive and A negative in the other case. The solutions of (18) are of course $e^{\sqrt{A}s}$ and $e^{-\sqrt{A}s}$, the first growing rapidly, and the second decreasing rapidly as $s \rightarrow \infty$, if $A > 0$. A glance at the integration by parts following (16) shows that $\text{Ai}(s)$ remains bounded when $s \rightarrow \infty$. So the comparison with $e^{\sqrt{A}s}$ must be dismissed, and we might reasonably guess that $\text{Ai}(s)$ is rapidly decreasing in this case. When $s < 0$ we take $A < 0$ in (18). The exponentials $e^{\sqrt{A}s}$ and $e^{-\sqrt{A}s}$ are now oscillating, and we can therefore presume that $\text{Ai}(s)$ should have an oscillatory character as $s \rightarrow -\infty$.

Theorem 3.1 *Suppose $u > 0$. Then as $u \rightarrow \infty$,*

$$(i) \quad \text{Ai}(-u) = \pi^{-1/2} u^{-1/4} \cos\left(\frac{2}{3}u^{3/2} - \frac{\pi}{4}\right)(1 + O(1/u^{3/4})).$$

$$(ii) \quad \text{Ai}(u) = \frac{1}{2\pi^{1/2}} u^{-1/4} e^{-\frac{2}{3}u^{3/2}} (1 + O(1/u^{3/4})).$$

To consider the first case, we make the change of variables $x \mapsto u^{1/2}x$ in the defining integral with $s = -u$. This gives

$$\text{Ai}(-u) = u^{1/2} I_-(u^{3/2}),$$

where

$$(19) \quad I_-(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{it(x^3/3-x)} dx.$$

Now write

$$I_-(s) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-s\Phi(x)} dx,$$

where $\Phi(x) = \Phi_-(x) = x^3/3 - x$, and we shall apply Proposition 2.2, which in this case, since s is purely imaginary, is the method of stationary phase. Note that $\Phi'(x) = x^2 - 1$, so there are two critical points, $x_0 = \pm 1$; observe that $\Phi''(x) = 2x$; also $\Phi(\pm 1) = \mp 2/3$.

We break up the range of integration in (19) into two intervals $[-2, 0]$ and $[0, 2]$ each containing one critical point, and two complementary integrals, $(-\infty, -2]$ and $[2, \infty)$.

Now we apply Proposition 2.2 to the interval $[0, 2]$ with $s = -it$, $x_0 = 1$, $\psi = 1/2\pi$, $\Phi(1) = -2/3$, $\Phi''(1) = 2$, and get a contribution of

$$\frac{1}{2\sqrt{\pi}} e^{-i\frac{2}{3}t} \left(\frac{1}{(-it)^{1/2}} + O\left(\frac{1}{|t|}\right) \right),$$

in view of (8). Similarly the integral over $[-2, 0]$ contributes

$$\frac{1}{2\sqrt{\pi}} e^{i\frac{2}{3}t} \left(\frac{1}{(it)^{1/2}} + O\left(\frac{1}{|t|}\right) \right).$$

Finally, consider the complementary integrals. The first is

$$\int_{-\infty}^{-2} e^{it\Phi(x)} dx = \lim_{N \rightarrow \infty} \int_{-N}^{-2} e^{it\Phi(x)} dx = \lim_{N \rightarrow \infty} \frac{1}{it} \int_{-N}^{-2} \frac{d}{dx} (e^{it\Phi(x)}) \frac{dx}{\Phi'(x)},$$

where $\Phi'(x) = x^2 - 1$. So an integration by parts shows that this is $O(1/|t|)$. The integral over $[2, \infty)$ is treated similarly. Adding these four contributions, and inserting them in the identity $\text{Ai}(-u) = u^{1/2} I_-(u^{3/2})$, proves conclusion (i) of the theorem.⁶

To deal with the conclusion (ii) of the theorem, we make the change of variables $x \mapsto u^{1/2}x$ in the integral (15), with $s = u$. This gives us, for $u > 0$,

$$\text{Ai}(u) = u^{1/2} I_+(u^{3/2}),$$

where

$$(20) \quad I_+(s) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-sF(x)} dx$$

and $F(x) = -i(x^3/3 + x)$. Now when $s \rightarrow \infty$, the integrand in (20) again oscillates rapidly, but here in distinction to the previous case, there is no critical point on the real axis, since the derivative of $x^3/3 + x$ does not vanish. A repeated integration by parts argument (such as we have used before) shows that actually the integral $I_+(s)$ has fast decay as $s \rightarrow \infty$. But what is the exact nature and order of this decrease? To answer this question, we would have to take into account the precise cancellations inherent in (20), and doing this by the above method does not seem feasible.

A better way is to follow the guiding principle used in the asymptotics of the Bessel function, and to deform the line of integration in (20) to a contour on which the imaginary part of $F(z)$ vanishes; having done this, one might then hope to apply Laplace's method, Proposition 2.1, to find the true asymptotic behavior of $I_+(s)$, as $s \rightarrow \infty$.

We describe the idea in the more general situation in which we assume only that $F(z)$ is holomorphic. To follow the approach suggested, we seek a contour Γ so that:

⁶An alternative derivation of this conclusion can be given as a consequence of the relation of the Airy function with the Bessel functions. See Problem 3 below.